

A Theory of International Boycotts

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Abstract

This paper provides the first systematic analysis of international boycotts within a neoclassical trade framework. A coalition of households maximizes its own welfare subject to keeping the welfare of a foreign country below a tolerated level. I establish a general equivalence result: equilibria with such a boycotting coalition coincide with equilibria in which atomistic, price-taking boycotters maximize utility subject to fixed expenditure budgets on each country's goods and, equivalently, net spending budgets on each country's factor services. Optimal boycotts (i) take the form of prohibitions on importing foreign goods unless foreign comparative advantage is sufficiently large, and (ii) lead boycotters to increase their labor supply to spend more on domestic goods when income effects outweigh substitution effects. A boycott by all domestic households is equivalent to targeted import tariffs and to a voluntary export restriction by foreign firms covering the boycotted goods, and the optimal boycott mirrors the uniform structure of optimal import tariffs; however, the effectiveness of boycotts is diminished by leakage effects from nonboycotting households. Thus, while boycotts share key similarities with conventional trade policy instruments, they introduce complexities that position them as distinct tools of geoeconomic influence.

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1 Introduction

In international trade and geopolitical relations, boycotts have emerged as a tool of market influence, akin in some respects to government-imposed sanctions and tariffs. While governments often employ sanctions and tariffs to exert political or economic pressure, boycotts reflect the collective choice of a coalition of households.

International boycotts raise several questions: What are boycotts in a neoclassical economic sense? How should members of a boycotting coalition allocate their consumption and labor supply to achieve their objective? How do international boycotts relate to and differ from government-imposed sanctions or import tariffs?

Historically, boycotts have been used in social (Weems, 1995) and political (Kam and Deichert, 2020) movements. Economists, however, have not given extensive attention to the economic consequences of boycotts. This paper aims to study the economics of international boycotts in a neoclassical trade model. To my knowledge, this is the first treatment of consumer boycotts using the workhorse frameworks of mainstream economics.

In my baseline model à la Dornbusch et al. (1977), firms in two countries, Home and Foreign, produce and export a continuous set of differentiated goods ranked in decreasing order of relative productivity or comparative advantage for the Home country. An exogenous share of domestic households forms a boycott coalition. The boycotting coalition behaves to maximize the welfare of its members, subject to holding Foreign households' welfare at a tolerated level, while internalizing the effect of its demand on equilibrium wages and prices. I define the corresponding notion of competitive equilibrium (CE) with a boycotter coalition and show, as my main result, that this internalization has a remarkably simple structure in equilibrium: the same allocations arise in a CE in which each boycotter is an atomistic price taker who maximizes utility *subject to fixed expenditure budgets on each country's goods*.

This equivalence result is general and applies beyond the DFS model to multi-country, multi-good settings with heterogeneous household economies that feature constant returns to scale (CRS) production and a single labor factor per country. Intuitively, under these conditions, equilibrium goods prices and production depend solely on wages, so that the factor content of any consumption bundle is linear in consumption; in the general model, each boycotting household's equilibrium behavior is characterized by net spending budgets on each country's factor services.

Therefore, boycotters ultimately need only choose, individually, the total budget they spend on each country's goods to achieve their goal of trading off their own welfare against the harm imposed on Foreign households. This choice can be analyzed from the perspectives of consumption and labor supply.

On the consumption side, boycotters allocate their demand across goods, given

fixed expenditures on each country’s goods. For any CE with boycotters having positive expenditure on domestic goods, the set of completely boycotted Foreign goods is nonempty. This result highlights that the usual notion of a boycott, a blanket prohibition of certain goods, is optimal to penalize Foreign welfare. In addition, every optimal boycott takes the form of a cutoff in the space of foreign comparative advantage; boycotters only import goods for which Foreign’s comparative advantage is sufficiently large.

The labor supply side of boycotts is equally important when labor is supplied elastically. A boycotter faces a higher price index than nonboycotters because she buys some goods domestically at less competitive prices. When income effects outweigh substitution effects, she optimally responds by working more than nonboycotters and by directing a disproportionate share of the additional earnings to domestic goods, thereby reinforcing the boycott; when substitution effects dominate, she optimally works less.

In relation to trade flows, expanding the set of boycotted foreign goods directly reduces imports by lowering the share of goods purchased abroad. It indirectly boosts domestic wages through increased demand for domestic goods. Although higher domestic wages encourage imports, the direct effect dominates, resulting in a net reduction in imports. Concurrently, the expanded boycott reduces the equilibrium cutoff for domestic goods exports, thereby decreasing exports to restore the trade balance. This mechanism highlights an equivalence between a comprehensive consumer boycott and a voluntary export restriction (VER) by Foreign firms covering the boycotted goods: refusing to buy and refusing to sell remove the same transactions from the Home market and are indistinguishable in equilibrium.

However, the presence of nonboycotting households reduces the effectiveness of a boycott through leakage effects. Specifically, the boycott indirectly raises domestic wages by shifting demand toward domestic goods, making foreign goods relatively cheaper for nonboycotters. Consequently, nonboycotters increase their imports from abroad, partially offsetting the boycott’s intended impact. This leakage, consistent with existing literature on optimal unilateral consumer actions motivated by altruistic preferences (see [Kaufmann et al. \(2024\)](#) and [Trammell \(2025\)](#)), weakens the boycott’s effectiveness. To maintain the same pressure on foreign welfare, boycotters thus need to target a broader set of goods as the share of nonboycotters increases.

In comparing full boycott coalitions with trade sanctions, a consumer boycott by all domestic households can be replicated by good-specific import tariffs. In particular, in a scenario where every household boycotts a range of foreign goods beyond the free-trade equilibrium cutoff, one can construct tariffs high enough to prevent imports of exactly those goods. Both the boycott and tariff scenarios result in identical equilibrium allocations and trade flows because they produce the same shifts in relative prices and

trade balance. Moreover, combining the results of [Becko \(2024\)](#) and [Opp \(2010\)](#), the optimal import tariffs in this setting are uniform, and the optimal boycott implements the same pattern of import exclusion as a uniform import tariff: both remove from imports the goods in which Foreign’s comparative advantage is smallest.

Comparing boycotts with linear import tariffs clarifies their equilibrium effects. Unlike boycotts, linear tariffs directly increase the cost of foreign goods, shifting the relative productivity curve and creating a set of nontraded goods. Boycotts, however, do not alter comparative advantage; rather, they operate through implicit income transfers from Foreign to Home due to geographic differences in household preferences. These transfers shift only the balanced trade curve, leading to higher domestic wages and reduced imports without creating nontraded goods when nonboycotting households are present, as they continue to trade all goods. Consequently, Foreign households lose access to efficiently produced Home goods, while domestic households benefit through improved terms of trade.

Extending the model highlights several important considerations regarding substitution elasticities and strategic retaliation. Higher substitution elasticities improve the effectiveness of boycotts, as boycotters find it easier to switch to alternative products. Additionally, considering retaliation introduces strategic interaction between countries, creating incentives akin to tariff wars.

1.1 Literature Review

This paper contributes to the economic literature by analyzing international boycotts through the lens of a neoclassical trade model, drawing on existing research on trade policies, economic sanctions, non-individualistic consumer behavior, and empirical studies of the trade effects of political conflict and international boycotts.

Trade Policies, Comparative Advantage, and Taste Shocks. The paper most closely related to mine is [Opp \(2010\)](#), which shows that the optimal import tariff in a DFS model is uniform, leading to a shift in imported varieties along the comparative advantage threshold and wage effects. My contribution is to derive analogous results on the optimal targeting of foreign varieties in a second-best scenario involving partial trade bans imposed by a subset of households.

[Costinot et al. \(2015\)](#) characterize optimal terms-of-trade manipulation in Ricardian settings, finding that optimal trade policies involve uniform tariffs on imports but heterogeneous taxes on exports. While their analysis encompasses more general preferences than the setting considered here, their problem is amenable to powerful public finance and Lagrangian techniques. In contrast, in my study of boycotts, which focuses on consumption sets and labor supply choices, Lagrangian techniques reach their limits

at the CE equivalence result for the general formulation of my problem. To further characterize the optimal set of boycotted goods, I rely on methods involving measurability conditions and proofs by contradiction. These techniques could be valuable in other studies of optimal policies without direct price instruments.

Additionally, participating in a boycott can be viewed as experiencing a preference shock favoring home-produced over foreign-produced goods for a specific subset of households. Although preference shocks have been extensively studied in international trade and macroeconomic literatures (see, e.g., [Stockman and Tesar \(1990\)](#), [Pavlova and Rigobon \(2007\)](#), and [Bai and Ríos-Rull \(2015\)](#)), our equivalence between equilibria with a boycotting coalition and equilibria with individual boycotters facing fixed expenditure budgets, as well as the optimal structure of targeted boycotts as prohibitions on specific goods, represent novel results within the context of my analysis.

Economic Sanctions. A burgeoning literature has emerged examining the economic impacts of sanctions and geopolitics in international trade (e.g., [Grossman and Helpman \(1994\)](#), [Antràs and i Miquel \(2011\)](#), [Traiberman and Rotemberg \(2023\)](#), [Clayton et al. \(2023\)](#); see also [Antràs \(2024\)](#) for a recent survey). In this literature, the closest paper to mine is [Becko \(2024\)](#), which connects trade sanctions to terms-of-trade manipulations. Similarly, [Itskhoki and Ribakova \(2024\)](#) analyze how various types of sanctions—including trade, financial, and payment restrictions—affect the Russian economy. Leveraging [Arkolakis et al. \(2012\)](#)’s welfare formulas, they find that the effect of trade sanctions can be measured by their impact on imports. In contrast, in my paper, boycotters engage in terms-of-trade manipulations, but nonboycotters’ responses to price adjustments can weaken the effectiveness of their actions.

Socially Responsible Consumers. This paper also contributes to the literature on market consequences of socially responsible behavior by price-taking consumers (e.g., [Dewatripont and Tirole \(2024\)](#), [Aghion et al. \(2023\)](#)). Related work by [Kaufmann et al. \(2024\)](#) and [Trammell \(2025\)](#) study optimal unilateral behavior by altruistic consumers who are not price-takers but rather internalize how their own actions will affect market outcomes in equilibrium. My modeling approach occupies an intermediate position between these two strands: individual boycotters remain price-takers but make their consumption and labor supply choices *among* (a constraint) those that hold the total expenditures of the boycott coalition on each country’s goods fixed. This approach allows me to show leakage effects driven by nonboycotters’ responses to equilibrium price changes while preserving the tractability of standard competitive equilibrium analysis.

International Boycotts and Political Conflict. This paper is motivated by empirical research on international boycotts and the broader literature documenting

the effects of political conflicts on trade patterns (Iyoha et al. (2024); Gopinath et al. (2025) show empirically that trade flows often align along geopolitical boundaries). Specifically, the "surrogate" international boycotts I study are driven not by firms' direct misconduct but by geopolitical tensions between countries, in which firms are targeted as representatives of their host countries. Prominent examples include the resurgent calls for boycotts of firms in and doing business with Israel during the Israel–Gaza war and the US boycott of French products following France's opposition to the Iraq war (Ashenfelter et al., 2007; Chavis and Leslie, 2009; Pandya and Venkatesan, 2016). Additional examples are the Arab League's boycotts of Danish products in 2005 and of French products in 2015 (Ammitzbøll and Vidino, 2007; Mirza et al., 2020). Moreover, government involvement can amplify international boycotts: Bohman and Pårup (2022) finds evidence that nearly one-third of all Chinese consumer boycotts have state support, suggesting that international boycotts that I study through a neoclassical framework serve as significant instruments of contemporary statecraft.

Outline. The rest of the article is organized as follows. Section 2 describes our baseline DFS model and presents all results in this setting. Section 3 generalizes our main result and discusses extensions of our analysis. Section 4 offers some concluding remarks. All formal proofs can be found in the Appendix.

2 A Benchmark DFS Model

I construct a theoretical framework based on the Dornbusch et al. (1977) model of international trade and study the role of consumer boycotts.

2.1 Environment

Two countries, namely, Home (H) and Foreign (F), engage in trade over a continuum of goods indexed by $j \in [0, 1]$.

Production: The model's production side is characterized by a single input, labor l , with a production function that exhibits a constant marginal product of labor for each good j . This marginal product is country specific, with $z_H(j)$ for Home and $z_F(j)$ for Foreign. Home's productivity in producing good j relative to Foreign's is given by $Z(j) = \frac{z_H(j)}{z_F(j)}$. Goods j are ordered so that Home's relative productivity is decreasing. Total labor endowments are denoted by l_H and l_F for Home and Foreign, respectively, and labor moves freely across the production of different goods j within a country. Markets are perfectly competitive, and labor is the sole determinant of production costs, leading to equilibrium wages w_H and w_F in Home and Foreign, respectively.

I distinguish between regular households in Home or Foreign, who do not participate in the boycott, and domestic boycotters, who boycott goods from Foreign.

Regular Households: Regular households in both countries are atomistic and derive utility from a diverse set of goods irrespective of provenance, and their preferences are represented by a Cobb–Douglas utility function, expressed as:

$$U(\mathbf{q}^i) = \int_0^1 s(j) \log(q^i(j)) dj \quad (1)$$

where $q^i(j)$ denotes the quantity of good j consumed by a regular household in country i and $s(j) > 0$ signifies the expenditure share on good j , which sums to 1 across all goods. I denote by $S(j) = \int_0^j s(j') dj'$ the fraction of income spent on goods with index value below j , with $S(1) = 1$. The budget constraint of regular households in country $i \in \{H, F\}$ is:

$$\int_0^1 p(j) q^i(j) dj = w_i l_i \quad (2)$$

where $p(j)$ is the equilibrium world price of good j .

Given wages and prices, regular households make demand choices to maximize their utility (1) subject to their budget constraint (2). Since each good is specific to a particular producer and there is a unique factor, labor, the non-substitution theorem – See (Samuelson, 1951; Stiglitz, 1970), and (Acemoglu and Azar, 2020; Toda and Walsh, 2024) for recent treatments – implies that prices are uniquely determined by the supply side of the economy $p(j; \{w_H, w_F\})$. Similarly, the agents' demands are unique given the prices. Therefore utilities of all nonboycotters are determined by the vector of world wages. Let $U^H(\{w_H, w_F\})$ and $U^F(\{w_H, w_F\})$ denote the utilities of regular households in Home and Foreign, respectively.

Boycotter Coalition: An exogenous share λ of domestic households forms a non-atomistic coalition of boycotters. They derive utility from the same Cobb–Douglas utility function (1). The coalition internalizes the effect of its choices on wages¹ and behaves so as to maximize welfare subject to maintaining a given welfare level v^F for Foreign households. Formally, following Kaufmann et al. (2024), we consider the following market structure. First, the boycotting coalition submits its demand for Home- and Foreign-produced goods $\mathbf{q}^B = \mathbf{q}_H^B + \mathbf{q}_F^B$. Then, the wages $w_H(\mathbf{q}^B), w_F(\mathbf{q}^B)$ are chosen

¹It is enough to internalize the effect on wages to internalize the effect on consumption goods prices $p(j, \mathbf{q})$ due to the non-substitution theorem that provides a mapping from wages to consumption goods prices $p(j; \{w_H, w_F\})$.

to clear the markets for domestic and foreign labor.² Finally, the equilibrium quantities are produced and consumed, generating a Foreign utility $U^F(\{w_H(\mathbf{q}^B), w_F(\mathbf{q}^B)\}) = v^F$.

The coalition correctly predicts the above outcomes and maximizes

$$U^B(v^F) = \max_{\mathbf{q}^B = \mathbf{q}_H^B + \mathbf{q}_F^B} \int_0^1 s(j) \log(q_H^B(j) + q_F^B(j)) dj \quad (3)$$

$$\text{s.t.} \int_0^1 p_H(j, \mathbf{q}^B) q_H^B(j) dj + \int_0^1 p_F(j, \mathbf{q}^B) q_F^B(j) dj = w_H(\mathbf{q}^B) l_H \quad (4)$$

$$U^F(\{w_H(\mathbf{q}^B), w_F(\mathbf{q}^B)\}) = v^F \quad (5)$$

Before defining the equilibrium, I introduce the price schedules that appear in the boycotters' problem. For each good j , let

$$p_H(j) = \frac{w_H}{z_H(j)} \quad \text{and} \quad p_F(j) = \frac{w_F}{z_F(j)} \quad (6)$$

denote the domestic prices at which good j can be purchased when produced in Home and Foreign, respectively. Under perfect competition, these prices equal marginal costs, and the world price of good j is $p(j) = \min\{p_H(j), p_F(j)\}$. The distinction matters for boycotters: by refusing to buy a Foreign good j that Home would not produce under free trade, boycotters create demand for its domestic production, albeit at the less competitive price $p_H(j) > p(j)$. I collect these prices in the schedules $\mathbf{p}_H = \{p_H(j)\}_{j \in [0,1]}$, $\mathbf{p}_F = \{p_F(j)\}_{j \in [0,1]}$, and $\mathbf{p} = \{p(j)\}_{j \in [0,1]}$. In the rest of this paper, I define a CE with boycotters as follows.

Definition 1 (Competitive Equilibrium with a Boycotter Coalition). *A CE with a boycotter coalition corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule of world prices $\mathbf{p}$, and domestic prices $\{\mathbf{p}_H, \mathbf{p}_F\}$, consumption schedules $\{\mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B\}$, foreign utility v^F , and output schedules that satisfy:*

- (i) *Firm optimization: price equal firms' marginal cost,*
- (ii) *Regular household utility maximization: regular households in Home and Foreign maximize (1) subject to (2),*
- (iii) *Boycotter coalition optimization: domestic boycotters maximize (3) subject to (4), and (5),*

²For demand schedules that exhaust the coalition's budget, market-clearing wages $w_H(\mathbf{q}^B), w_F(\mathbf{q}^B)$ exist and are unique up to a normalization: normalizing $w_F = 1$, Foreign labor market clearing pins down $\omega = w_H/w_F$ uniquely from the coalition's expenditure on Foreign goods by the strict monotonicity established in Appendix A.1 (Equation (35)), and Home labor market clearing then follows from Walras's law. The non-substitution theorem maps these wages to goods prices.

(iv) Boycotter coalition choices are consistent with equilibrium wages:

$$\{w_H(\mathbf{q}^{*B}), w_F(\mathbf{q}^{*B})\} = \{w_H, w_F\} \quad (7)$$

(v) Labor market clearing in each country, where regular households purchase each good from its least-cost producer while boycotters' demands are charged to each country they purchase from:

$$\int_0^1 \frac{((1-\lambda)q^H(j) + q^F(j))\mathbf{1}\{p_H(j) \leq p_F(j)\} + \lambda q_H^B(j)}{z_H(j)} dj = l_H \quad (8)$$

$$\int_0^1 \frac{((1-\lambda)q^H(j) + q^F(j))\mathbf{1}\{p_H(j) > p_F(j)\} + \lambda q_F^B(j)}{z_F(j)} dj = l_F \quad (9)$$

This definition deserves some explanation. Conditions (i), (ii), and (v) are the standard [Dornbusch et al. \(1977\)](#) equilibrium conditions: firms price at marginal cost, regular households in both countries maximize utility taking wages and prices as given, and labor markets clear in each country. Conditions (iii) and (iv) embed the boycotter coalition's behavior. In contrast to regular households, the coalition is non-atomistic: in condition (iii), it chooses its demand schedule to maximize the utility of its representative member subject to its budget constraint and to holding Foreign households' utility at the tolerated level v^F , internalizing that its demand moves the market-clearing wages and, through the non-substitution theorem, all goods prices. Condition (iv) is the consistency requirement that the wages the coalition internalizes when choosing its demand $\mathbf{q}^{*B}$ coincide with the equilibrium wages. The definition thus makes the coalition a large player in an otherwise competitive economy, as in [Kaufmann et al. \(2024\)](#). I next show that this internalization has, in equilibrium, a remarkably simple structure: the same allocations arise in a CE in which each boycotter is an atomistic price taker who maximizes utility subject to her budget constraint and to a fixed expenditure on each country's goods.

2.2 An Equivalence Result

I introduce some notation to characterize CEs with a boycotter coalition further. Let $E(\mathbf{q}_i^B) \equiv \lambda \int_0^1 p_i(j) q_i^B(j) dj$, for $i \in \{H, F\}$, denote the boycotter coalition's total expenditure on Home and Foreign goods, respectively, valued at the domestic prices $p_i(j)$ defined in (6), which can be distinct from world prices.

Individual Boycotters: I now consider an alternative setting in which an exogenous share λ of atomistic domestic households derives utility from the same Cobb–Douglas utility function (1). Given wages and prices, they maximize their utility sub-

ject to spending a budget $b_F \in [0, w_H l_H]$ on Foreign goods and $w_H l_H - b_F$ on domestic goods. Formally, the individual boycotters' problem is:

$$U^B(\{w_H, w_F, b_F\}) = \max_{\mathbf{q}^B = \mathbf{q}_H^B + \mathbf{q}_F^B} \int_0^1 s(j) \log(q_H^B(j) + q_F^B(j)) dj \quad (10)$$

$$\text{s.t: } \int_0^1 p_F(j) q_F^B(j) dj = b_F \quad (11)$$

$$\int_0^1 p_H(j) q_H^B(j) dj = w_H l_H - b_F \quad (12)$$

In this alternative setting, boycotters make individually optimal consumption choices while keeping their overall expenditure on Home and Foreign goods fixed. The following definition formalizes the notion of CE under this alternative choice set for boycotters.

Definition 2 (Competitive Equilibrium with Individual Boycotters with Fixed Budgets). *A CE with boycotters corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule of world prices $\mathbf{p}$, and domestic prices $\{\mathbf{p}_H, \mathbf{p}_F\}$, consumption schedules $\{\mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B\}$, budget for foreign goods b_F , and output schedules that satisfy:*

1. *Firm optimization: price equal firms' marginal cost,*
2. *Regular household utility maximization: regular households in Home and Foreign maximize (1) subject to (2),*
3. *Individual boycotters' utility maximization: domestic boycotters take wages and prices as given and maximize (10) subject to (12), and (11),*
4. *Labor market clearing in each country (8) and (9).*

The following proposition provides an equivalence between the two notions of CE: one in which a coalition of boycotters internalizes its effect on wages while holding Foreign welfare at v^F , and one in which atomistic boycotters take prices as given and hold their overall expenditure on each country's goods fixed. For this result, I maintain the assumptions of an interior trade equilibrium, $\bar{j} \in (0, 1)$, with Z continuously differentiable and strictly decreasing, and I consider tolerated welfare levels v^F in the attainable range of Foreign welfare.

Proposition 1 (Equivalence of CE with a Boycotter Coalition to CE with Individual Boycotters with Fixed Budgets). *A tuple $\{w_H, w_F, \mathbf{p}, \mathbf{p}_H, \mathbf{p}_F, \mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B, v^F\}$ is a CE with a boycotter coalition if and only if there exists $b_F(v^F)$ such that $\{w_H, w_F, \mathbf{p}, \mathbf{p}_H, \mathbf{p}_F, \mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B, b_F(v^F)\}$ is a CE with individual boycotters with fixed budgets.*

This proposition states that the coalition's internalization of wages reduces, in equilibrium, to a single choice: how to split total spending between Home and Foreign

goods. Given the tolerated level of Foreign welfare v^F , there is a unique expenditure budget $b_F(v^F)$ on Foreign goods such that atomistic, price-taking boycotters who maximize utility subject to this budget replicate the coalition's optimum. The proposition is general, as I show in Section 3: boycotters ultimately need only choose the overall budget they allocate to each country's goods (and equivalently, to each country's factor services) and maximize their welfare given those budgets. They can do so in two ways: first, by optimally allocating their consumption between Home and Foreign goods within their budgets; and second, by expanding their budgets through their labor-supply choice when labor supply is endogenous. Next, I analyze the consumption side of boycotts in this setting with a fixed labor supply.

2.3 The Consumption Side of Boycotts

Let $S(\mathbf{q}_i^B) \equiv \int_0^1 p_i(j)q_i^B(j)dj / (w_H l_H)$ denote the total expenditure, as a share of domestic income, on Home and Foreign goods by a representative boycotter, respectively.

Lemma 1 (Set of Boycotted Goods). *In any CE with individual boycotters with fixed budgets in which boycotters' expenditure on Home goods is positive, $b_F < w_H l_H$, the set of completely boycotted Foreign goods $\mathcal{J}_B = \{j \in [0, 1]; q_F^B(j) = 0\}$ is nonempty.*

This lemma implies that there is a complete prohibition on buying certain Foreign goods. More generally, the proof in the Appendix shows that boycotters choose two disjoint sets of Home and Foreign goods in which they will choose positive Home goods consumption or positive Foreign goods consumption.

In this setting, just as in the Dornbusch et al. (1977) economy, the patterns of trade for regular households are determined by a cutoff $\bar{j} = \bar{j}(\{w_H, w_F\})$ above which Foreign goods are imported, and let $S(\bar{j})$ denote the expenditure share of regular households on domestic goods. I further characterize the boycotted set of goods in the lemma below.

Proposition 2 (Optimal Set of Boycotted Goods). *Let $\mathcal{J}_B$ be the optimal targeted set of boycotted goods given a level of tolerated Foreign welfare v^F :*

1. *There exists no set of positive measure between $\mathcal{J}_B$ and the trade cutoff $\bar{j}$, and $\mathcal{J}_B$ is a convex set.*
2. *There exists $j_B \in [\bar{j}, 1]$ so that $\mathcal{J}_B = [0, j_B]$ (almost everywhere) boycotts goods up to j_B .*

The proof of this proposition highlights what matters for the optimal set of goods to boycott. Domestic income from the boycott is proportional to the boycotters' expenditure share on Home-sourced goods, $S(\mathbf{q}_H^B) = 1 - b_F/(w_H l_H)$, which is pinned down by the budget split. When the boycott constraint binds, the boycotters' Cobb–Douglas

expenditure shares renormalize within each sourcing set (see Appendix A.2), so $S(\mathbf{q}_H^B)$ exceeds the preference mass $\int_{\mathcal{J}_B} s(j) dj$ of the boycotted set; the two coincide when the constraint does not bind. This expenditure share is the only determinant of the trade cutoff and the Foreign wage because of the balanced trade equation, which equalizes domestic income with expenditure on Home-produced goods:

$$w_H l_H = \underbrace{(1 - \lambda) S(\bar{j}) w_H l_H}_{\text{regular domestic households}} + \underbrace{\lambda S(\mathbf{q}_H^B) w_H l_H}_{\text{domestic boycotters}} + \underbrace{S(\bar{j}) w_F l_F}_{\text{Foreign households}} \quad (13)$$

Because the balanced trade equation depends on the boycott only through the budget split, any set of boycotted goods consistent with the same budgets $(w_H l_H - b_F, b_F)$ yields the same relative wage and trade cutoff and, hence, the same Foreign welfare. Among all such sets, the boycotters' own optimality selects the goods closest to the trade cutoff: the first-order conditions in Appendix A.2 assign each good to the source with the lower shadow cost and, since the domestic price premium $p_H(j)/p_F(j) = \omega/Z(j)$ is strictly increasing in j , the Home-sourced set is an interval $[0, j_B]$. Goods of lower index j are cheaper for Home to produce, so sourcing them domestically minimizes the cost of the boycott to its members while leaving Foreign's welfare unchanged. Thus, for any given utility v^F for Foreign, the optimal targeted set of goods to boycott includes those closest to the trade cutoff.

2.4 The Labor Supply Side of Boycotts

I now turn to the labor supply side of boycotts. In our baseline DFS model with log utility of consumption, the income and substitution effects offset each other, so labor supply, even if endogenous, does not respond to wage changes.

I modify the DFS setup so that households choose consumption goods $\mathbf{q}$ and labor supply l_i to maximize

$$U(\mathbf{q}^i, l_i) = \int_0^1 s(j) u(q^i(j)) dj - v(l_i) \quad (14)$$

We consider CRRA utility for goods and an isoelastic disutility of labor:

$$u(q) = \frac{q^{1-\gamma} - 1}{1-\gamma} \quad \text{and} \quad v(l) = \frac{l^{1+\eta}}{1+\eta}, \quad (15)$$

with $1 \neq \gamma > 0$ and $\eta > 0$.

We can solve the boycotters' problem in two steps: first, they choose the consumption bundle of goods given their income, and then they choose their labor supply. In the first step, Propositions 1 (equivalence of the CE with a boycotter coalition to the CE with individual boycotters with fixed budgets) and 2 (optimal boycott set is a cut-

off) generalize to these preferences, so we focus on the labor supply aspect of boycotts. For this margin, I evaluate the boycotter's problem in its prohibition form: facing the equilibrium boycott cutoff j_B , the boycotter buys goods in $[0, j_B]$ domestically at prices $p_H(j)$ and the remaining goods at world prices, subject to a single budget constraint. This formulation holds the boycotted set fixed as the labor supply varies and yields a transparent comparison with regular households using a single effective price index.

Proposition 3 (Labor Supply of Boycotters). *In every optimal boycott, boycotters work more than regular Home households if and only if income effects outweigh substitution effects: $l^{B*} > l_H^* \iff \gamma > 1$.*

To arrive at this proposition, observe that households' labor supply choice in the second step is determined by the indirect utility

$$V(w_i, l_i) = \frac{\left(\frac{w_i l_i}{P_i}\right)^{1-\gamma} - 1}{1-\gamma} - \frac{l_i^{1+\eta}}{1+\eta}, \quad (16)$$

where $P_i \equiv \left(\int_0^1 s(j)^{\frac{1}{\gamma}} p_i(j)^{\frac{\gamma-1}{\gamma}} dj\right)^{\frac{\gamma}{\gamma-1}}$ is the ideal price index faced by the household, which is increasing in each price $p_i(j)$ (see Appendix A.4). The optimal labor supply is therefore

$$l_i^* = \left(\frac{w_i}{P_i}\right)^{\frac{1-\gamma}{\eta+\gamma}}.$$

A boycotter faces a penalty in her consumption basket because she must pay more for the goods she buys domestically, so her price index exceeds that of a regular Home household, $P_B > P_H$. The comparison of labor supplies is then governed by the strength of income effects relative to substitution effects:

$$\frac{l^{B*}}{l_H^*} = \left(\frac{P_H}{P_B}\right)^{\frac{1-\gamma}{\eta+\gamma}} > 1 \iff \gamma > 1.$$

When $\gamma > 1$, income effects dominate: to offset the cost of buying domestically produced goods, the boycotter works more. In doing so, she earns more income; since her consumption is financed entirely by labor earnings and the goods in $[0, j_B]$ are sourced domestically, a disproportionate share of her increased income is spent on domestic goods. In equilibrium, this is reinforced by the fact that the coalition's spending on Foreign goods is pinned down in level, at $b_F(v^F)$, by the tolerated Foreign welfare (Proposition 1), so the labor supply margin ultimately finances additional spending on domestic goods. This additional spending on domestic varieties helps tilt the relative price in favor of Home production. In equilibrium, this upward pressure on domestic spending raises the domestic wage (and, through the cutoff condition that pins down production, lowers the trade cutoff $\bar{j}$ relative to the free-trade result), thereby harming

Foreign's utility. When $\gamma < 1$, substitution effects dominate, and the boycotter responds to the higher cost of her consumption basket by working less; the labor-supply margin then dampens rather than reinforces the boycott. Consistent with the baseline model, the two effects exactly offset in the log case, where boycotters and regular households supply the same amount of labor.

2.5 Comparison with Trade Sanctions

Full Boycotts as Targeted Import Tariffs. To compare boycotts with sanctions, I first analyze the extreme cases of an economy where all Home households are boycotters and an economy with good-specific tariffs.

Consider our DFS economy when every domestic household is a boycotter ($\lambda = 1$). Start with the no-boycott equilibrium, where j_B is the free-trade cutoff, denoted $\bar{j}_{NB}$. Now, introduce a boycott $\mathcal{J}_B = [0, j_B]$, with cumulative expenditure share $S([0, j_B]) < 1$ and $j_B > \bar{j}_{NB}$.

In a DFS economy without boycotters ($\lambda = 0$), we can then construct good-specific import tariffs on goods indexed by $j \in (\bar{j}(j_B), j_B]$, which replicate the equilibrium allocations (and trade flows) of the DFS economy with a boycott. The idea is to set good-specific tariffs high enough that no goods in $(\bar{j}(j_B), j_B]$ are imported. Note that the tariffed range extends down to the new export cutoff $\bar{j}(j_B) < \bar{j}_{NB}$, not merely to the free-trade cutoff: at the higher equilibrium relative wage, Home households would otherwise import all goods above $\bar{j}(j_B)$. Since no tariffs are collected in equilibrium, the shift in the trade balance (and export cutoff $\bar{j}(j_B)$) in the boycott economy and good-specific tariff economy are the same, and both arise from Home households allocating more toward Home goods. Because the BT condition for the full-boycott-coalition economy pins down the equilibrium relative wage,

$$\omega = \frac{S(\bar{j}(j_B))}{1 - S(j_B)} \cdot \frac{l_F}{l_H}, \quad (17)$$

the good-specific tariffs are constructed such that the tariff-adjusted relative productivity curve lies above this relative wage on the excluded range and crosses it at j_B :

$$(1 + t(j))Z(j) > \omega \text{ for all } j \in (\bar{j}(j_B), j_B) \text{ and } (1 + t(j_B))Z(j_B) = \omega. \quad (18)$$

Therefore, each boycott is equivalent to goods-specific sanctions. In general, it is well known that any feasible allocation can be attained by good-specific and household-specific taxes. Thus, we need not limit ourselves to a comparison with goods-specific tariffs alone to understand the link between boycotts and sanctions.

Optimal Boycotts and Uniform Import Tariffs. What do *optimal* import tariffs look like in this setting? [Becko \(2024\)](#) shows that the optimal sanctions for reducing Foreign welfare at the least cost to Home have the same structure as terms-of-trade-maximizing tariffs but are larger in magnitude, and [Opp \(2010\)](#) shows that, when export taxes are unavailable, the terms-of-trade-maximizing import tariff in the DFS model is uniform. Taken together, these observations imply that the optimal import tariffs in this setting are uniform. This is consistent with my characterization of the optimal boycott in Proposition 2: a uniform import tariff removes from imports precisely the goods in which Foreign’s comparative advantage is smallest (those adjacent to the trade cutoff), and the optimal boycott concentrates its prohibition on exactly the same goods. The optimal boycott therefore implements the same pattern of import exclusion and, hence, the same equilibrium quantities as a uniform import tariff.

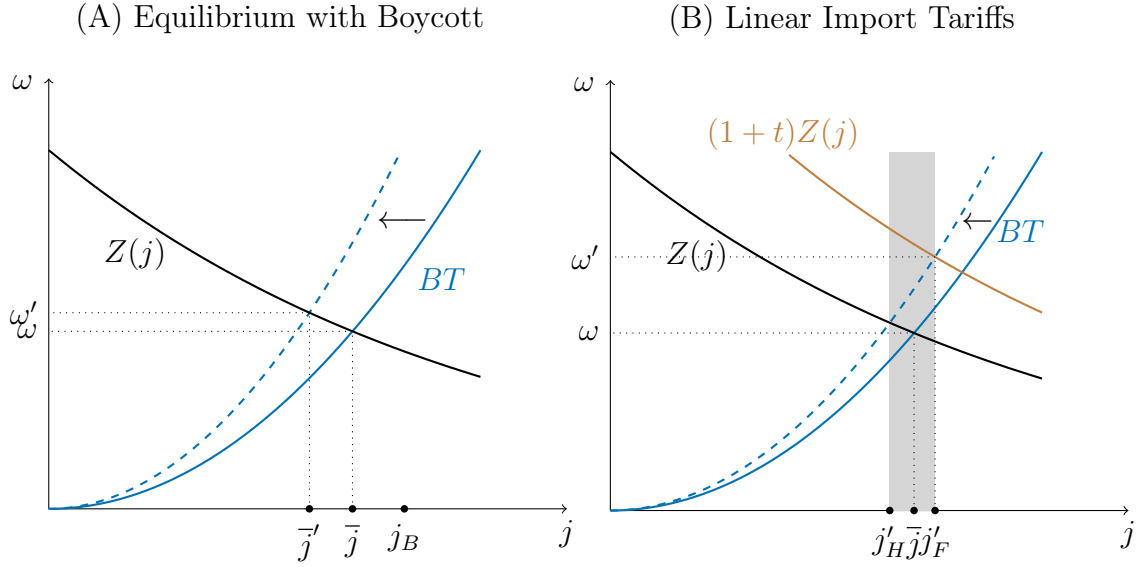
I now compare in more detail the equilibrium with a boycott to the equilibrium with a linear import tariff.³ We can think of sanctions in the model with only regular domestic and Foreign households (no boycotters) as a tariff t on Foreign exports q_F such that the domestic price of Foreign goods is $p_H(j) = p(j)(1 + t \cdot q_F)$ and revenues are rebated lump-sum to domestic households. Figure 1 illustrates the equilibria with a boycott and with sanctions.

First, in Panel B, sanctions make Foreign exports more expensive and shift the relative productivity curve (for Foreign exports) according to $\omega = (1 + t)Z(j)$. At the same time, boycotts keep the relative productivity curve unchanged. Since tariffs are rebated to domestic households, this shifts the balanced trade curve slightly to the left. These two forces on net lead to an increase in the relative wage of Home households and a decrease in the range of goods imported from Foreign for sanctions (in contrast to the increase in that range in the case of boycotts). In this DFS economy, the sanctions equilibrium features a set of nontraded goods $[j'_H, j'_F]$, while the full-boycott equilibrium features a shift from the trade cutoff and relative wage from $(\bar{j}, \omega)$ to $(\bar{j}', \omega')$. Under a full boycott, the goods in $(\bar{j}', j_B]$ are produced in both countries for their respective households and are thus de facto nontraded; the distinction is that the boycott leaves the relative productivity curve unshifted, whereas the tariff rotates it. With nonboycotting households present ($\lambda < 1$), every good remains traded.

The correct intuition for consumer boycotts, however, is the economics of transfers when tastes differ across geographic regions. The boycotters’ income that would have been allocated to Foreign exports $\lambda w_H l_H [1 - S(\bar{j})]$ can be interpreted as a unilateral transfer from Foreign to Home. Now, since these transfers would be consumed on domestic goods due to the difference in tastes between boycotters and Foreign households,

³In general, I can treat sanctions as nonlinear tariffs $t(q_F)$ that capture a mixture of taxes and quotas, and show that equilibrium can be implemented with a linear tax under complete information.

Figure 1: Equilibrium effects of an increase in the boycott cutoff and an increase in import sanctions



Notes: An increase in boycott cutoff j_B (left) and an increase in import t sanctions (right). An increase in the boycott cutoff shifts the BT curve to the left, and the equilibrium trade cutoff $\bar{j}$ (relative wage of Home workers ω) decreases (increases) to equal $\bar{j}'$ (ω'). An increase in import tariffs t shifts the relative productivity curve for imports to the right, and the transfers of collected tariffs lead to a slight leftward shift of the BT curve. Home exports goods $[0, j_H']$, and Foreign exports $[j_F', 1]$. The equilibrium relative wage of Home workers is higher; however, there is no trade in the shaded region.

this shifts the BT curve (and only the BT curve) to the left.

The loss for Foreign households stems from their inability to obtain goods previously imported from Home that Foreign workers are less productive at making. Even if, in a multicountry model, the Foreign country could switch to the second-best producer of those goods, it would still be worse off as a result of the boycott. The gain for domestic regular households stems from the fact that trade is balanced. In a multicountry model, if trade were imbalanced and boycotters had to switch to the rest of the world rather than Home producers, then Home households would not be better off from the boycott.

Producer Boycotts and Voluntary Export Restrictions. The analysis so far considers consumer boycotts, which restrict purchases of Foreign goods on the demand side of the Home market. The same transactions can instead be restricted on the supply side: in a voluntary export restriction (VER), Foreign firms withhold their exports from the Home market. The accounting below shows that boycotts compress imports and exports together and makes the connection between consumer boycotts and VERs precise.

In the full-boycott-coalition economy, imports of foreign goods are

$$M_H = [1 - S(\mathcal{J}_b)]w_H l_H. \quad (19)$$

Thus, an expansion of the set of boycotted goods $\mathcal{J}_b$ directly decreases the share $1 - S(\mathcal{J}_b)$ and indirectly increases domestic wages due to increased demand for domestic goods. On net, the import share term dominates, and imports decrease.

Exports of domestic firm goods are

$$X_H = S(\bar{j}(j_B))w_F l_F. \quad (20)$$

An increase in the boycott cutoff j_B reduces the trade equilibrium cutoff $\bar{j}(j_B)$ and therefore decreases Home exports, which restores trade balance. Boycotts, therefore, distort both imports and exports. To make the connection with VERs precise, consider an economy A where everyone is a boycotter ($\lambda = 1$) and the set of boycotted goods is $\mathcal{J}_B = [0, j_B]$, and an economy B where Foreign firms voluntarily refrain from exporting to Home the goods in $[0, j_B]$ —equivalently, the would-be imports in $(\bar{j}(j_B), j_B]$. Then, economies A and B are equivalent: the two restrictions remove the same transactions from Home households' feasible set—in economy A, Home households refuse to buy precisely what, in economy B, Foreign firms refuse to sell—so Home households face the same effective price schedule, Foreign households are unconstrained in both economies, and the competitive equilibria coincide in all consumption allocations, production patterns, and wages. That is, a blanket boycott by all households is equivalent to a VER

by Foreign firms covering the boycotted goods: in equilibrium, demand-side refusal and supply-side withdrawal of the same transactions are indistinguishable.

A different producer-side instrument is a boycott by *Home* firms, which would refrain from selling to Foreign. Just as my focus on consumer boycotts is analogous to the restriction in [Opp's \(2010\)](#) to import tariffs alone, such Home producer boycotts would play the role of the export taxes studied by [Costinot et al. \(2015\)](#), who find that, in the DFS model, optimal import taxes are uniform while optimal export taxes are not. I do not characterize optimal Home producer boycotts here; note, however, that they are not interchangeable with consumer boycotts: by withdrawing export demand for Home labor without generating scarcity rents (prices equal unit labor costs in this Ricardian setting), a Home-side export restraint *lowers* Home's relative wage, whereas the consumer boycott raises it.⁴

2.6 Leakage from Nonboycotters

The equivalence between boycotts and trade sanctions developed above relies on all Home households participating in the boycott. The first important limitation of this connection is leakage: when only a share $\lambda < 1$ of households boycotts, the responses of nonboycotting households partially undo the boycott. With the presence of nonboycotters $\lambda < 1$, a boycott $\mathcal{J}_B = [0, j_B]$ leads to the imports from nonboycotters equal to

$$M_{NB} = (1 - \lambda)[1 - S(\bar{j}(j_b))]w_H l_H. \quad (21)$$

The boycott indirectly raises domestic wages by reallocating demand toward domestic goods. In addition, we saw in (20) that nonboycotters' share of spending on Foreign-produced goods, $1 - S(\bar{j}(j_b))$, increases when a boycott expands. The combination of the effect of boycotts on domestic wages and on this import share of nonboycotters raises the imports of nonboycotters, M_{NB} , from the foreign country. Therefore, a boycotter's unilateral attempts to prevent Foreign imports can be undermined by other households' responses to the wage and price changes she induces. This leakage is present in the literature on optimal unilateral behavior by consumers motivated by altruistic, non-individualistic preferences (see [Kaufmann et al. \(2024\)](#) and [Trammell \(2025\)](#)). This equilibrium leakage pushes boycotters to boycott a larger set of goods to attain the same level of Foreign welfare v^F : the required cutoff $j_B(\lambda, v^F)$ falls as the boycotting share λ rises, while nonboycotters' imports M_{NB} shrink with the nonboycotting share $(1 - \lambda)$.

⁴With Home exports restricted to $[0, j_R]$, $j_R < \bar{j}_{NB}$, the trade balance reads $\omega = \frac{S(j_R)}{1 - S(\bar{Z}^{-1}(\omega))} \cdot \frac{l_F}{l_H}$, since Home households remain unconstrained and import all goods above $\bar{Z}^{-1}(\omega)$. If $\omega \geq \omega_{NB}$, the right-hand side is strictly smaller than ω_{NB} , a contradiction; hence $\omega < \omega_{NB}$.

3 Generalization and Discussion

3.1 A General Equivalence Result

I now prove our result in its most general setup. There are countries $i \in \mathcal{I}$, goods $g \in \mathcal{G}$, and households $h \in \mathcal{H}_i$ in each country i . Within one specific country i^* , a subset of households $B_{i^*} \subset \mathcal{H}_{i^*}$ are boycotters. All remaining households are regular households. There are no trade taxes, so all households face the same world goods prices $p = \{p_g\}_{g \in \mathcal{G}}$ but earn local wages w_i in their country of residence.

Production: There is a single labor factor in each country, elastically supplied by households. In each country i and for each good $g \in \mathcal{G}$ there is a representative competitive firm operating a technology

$$F_{ig} : \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{G}|} \rightarrow \mathbb{R}_+, \quad y_{ig} \leq F_{ig}(\ell_{ig}, m_{ig}),$$

where ℓ_{ig} is the amount of country- i labor used in sector g and $m_{ig} = \{m_{i,jg}\}_{j \in \mathcal{G}}$ is a vector of inputs (quantities $m_{i,jg}$ of each good j used in producing good g in country i). Let $y_i = \{y_{ig}\}_{g \in \mathcal{G}}$ denote the vector of gross outputs of country i and $\ell_i = \sum_{g \in \mathcal{G}} \ell_{ig}$ its total labor demand. The representative firm in country i chooses outputs and inputs to maximize profits:

$$\begin{aligned} (\{y_{ig}\}_g, \{\ell_{ig}\}_g, \{m_{ig}\}_g) &\in \arg \max_{y, \ell, m} \sum_{g \in \mathcal{G}} [p_g y_{ig} - w_i \ell_{ig} - p \cdot m_{ig}] \\ \text{s.t. } y_{ig} &\leq F_{ig}(\ell_{ig}, m_{ig}) \quad \forall g \in \mathcal{G}. \end{aligned} \quad (22)$$

Global goods markets and local labor markets clear. Gross output of each good equals aggregate final consumption plus aggregate intermediate input use, and each country's labor demand equals the sum of its households' labor supplies:

$$\sum_{i \in \mathcal{I}} y_{ig} = \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}_i} c_g(h) + \sum_{i \in \mathcal{I}} \sum_{g' \in \mathcal{G}} m_{i,gg'} \quad \text{for all } g \in \mathcal{G}, \quad \ell_i = \sum_{h \in \mathcal{H}_i} \ell_i(h) \quad \text{for all } i \in \mathcal{I}. \quad (23)$$

Regular Households: Each regular household h in country i chooses a labor supply $\ell_i(h)$ and a vector of consumption $c(h) = \{c_g(h)\}_{g \in \mathcal{G}}$ to maximize utility subject to a budget constraint:

$$\ell_i(h), c(h) \in \arg \max_{\ell, c} u(c, \ell; h) \quad \text{s.t. } p \cdot c \leq w_i \ell. \quad (24)$$

Boycotter Coalition: Let $B_i \subseteq \mathcal{H}_i$ be the set of boycotting households in country i . The coalition internalizes the effect of its choices on wages and behaves so as to implement a point on their Pareto frontier subject to maintaining given welfare levels of all other households. For example, they may wish to ensure a sufficiently low level of welfare in a country they wish to punish for some geopolitical action. I rule out transfers among members of the boycotting coalition, but otherwise take no stance on how its members seek to redistribute among themselves. The no-transfer restriction is formalized by requiring each boycotting household to spend exactly its own wage income:

$$p \cdot c(h) = w_{i^*} \ell_{i^*}(h) \quad \text{for all } h \in B_{i^*}. \quad (25)$$

I formally integrate the boycotting coalition's problem by defining a competitive equilibrium in this setting.

Definition 3 (General Competitive Equilibrium with a Boycotter Coalition). *A CE with a boycotter coalition corresponds to a profile of world prices and wages $\{\{p_g\}, \{w_i\}\}$ and quantities for all households and firms $\{c(h), \ell_{i(h)}(h)\}_h, \{y_i, \ell_i\}_i$ such that it satisfies equations (22), (23), (24), and (25), and no other profile $\{\{p'_g\}, \{w'_i\}, \{c'(h), \ell'_{i(h)}(h)\}_h, \{y'_i, \ell'_i\}_i\}$ satisfies:*

- (i) *Equations (22), (23), (24), and (25) hold.*
- (ii) *The profile of utilities of households outside the boycotting coalition is the same as in the first profile, i.e.*

$$\{u(c(h), \ell_{i(h)}(h); h)\}_{h \notin B_{i^*}} = \{u(c'(h), \ell'_{i(h)}(h); h)\}_{h \notin B_{i^*}}.$$

- (iii) *The profile of utilities of households in the boycotting coalition Pareto dominates their baseline profile, i.e.*

$$\{u(c(h), \ell_{i^*}(h); h)\}_{h \in B_{i^*}} <_{\text{Pareto}} \{u(c'(h), \ell'_{i^*}(h); h)\}_{h \in B_{i^*}}.$$

This definition deserves some explanation. A CE with a boycotter coalition, which I also refer to as a *boycotting equilibrium*, is a profile of prices, wages, and allocations that satisfies the standard competitive conditions of profit maximization by firms (22), market clearing (23), utility maximization by regular households (24), and budget balance for boycotters (25) and at which the boycotting coalition implements a point on its Pareto frontier subject to preserving the utilities of all households outside the coalition. Conditions (i)–(iii) formalize the latter requirement: there must be no other profile consistent with competitive behavior by firms and regular households (condition (i)) that leaves every household outside the coalition exactly as well off (condition (ii)) while

making every boycotter weakly better off and at least one boycotter strictly better off (condition (iii)). The definition thus takes no stance on how the coalition redistributes welfare among its members (transfers among them are ruled out) and only requires that the coalition exhaust the gains available to it. It nests the coalition equilibrium of Section 2: there, the coalition has a representative member, and holding the representative Foreign household at the tolerated welfare level v^F pins down relative wages and, with them, the utility of every regular household.

The key technical step in generalizing Proposition 1 is to show that, in this class of economies, the factor content of any consumption bundle is well defined and linear in consumption. This is where the non-substitution theorem comes in (Samuelson, 1951; Stiglitz, 1970): with a single primary factor per country, constant returns to scale, and no joint production, equilibrium prices of goods and cost-minimizing quantities depend only on wages, not on the composition of demand. I first state precisely the assumptions on technology under which I invoke the theorem.

Assumption 1 (Non-substitution technology). *For each country $i \in \mathcal{I}$ and good $g \in \mathcal{G}$, production is carried out by a competitive firm operating a production function*

$$F_{ig} : \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{G}|} \rightarrow \mathbb{R}_+, \quad y_{ig} \leq F_{ig}(\ell_{ig}, m_{ig}),$$

where ℓ_{ig} is country i 's labor input and $m_{ig} \in \mathbb{R}_+^{|\mathcal{G}|}$ is a vector of intermediate-goods inputs.

The technology satisfies:

1. F_{ig} is continuous, strictly increasing in all inputs, and strictly quasi-concave.
2. F_{ig} is constant returns to scale: $F_{ig}(\lambda \ell, \lambda m) = \lambda F_{ig}(\ell, m)$ for all $\lambda > 0$.
3. There is no joint production: sector (i, g) only produces good g .
4. For any strictly positive factor price $w_i > 0$ and goods-price vector $p \in \mathbb{R}_{++}^{|\mathcal{G}|}$, the unit-cost problem

$$c_{ig}(w_i, p) \equiv \min_{\ell_{ig}, m_{ig}} \{ w_i \ell_{ig} + p \cdot m_{ig} : F_{ig}(\ell_{ig}, m_{ig}) \geq 1 \}$$

has a unique solution with strictly positive inputs and a differentiable unit-cost function.

5. For every strictly positive wage vector w , there exists a unique strictly positive goods-price vector $p(w)$ satisfying the zero-profit pricing equations $p_g = \min_{i \in \mathcal{I}} c_{ig}(w_i, p(w))$ for all $g \in \mathcal{G}$.⁵

⁵Existence and uniqueness of $p(w)$ hold under standard productiveness conditions on the world technology; see Samuelson (1951); Stiglitz (1970) and, for modern treatments, Acemoglu and Azar (2020) and Toda and Walsh (2024).

6. (Unique specialization and productiveness.) For every strictly positive wage vector w and every good g , the cost-minimizing producing country $i(g; w) \in \arg \min_{i \in \mathcal{I}} c_{ig}(w_i, p(w))$ is unique,⁶ and the equilibrium input–output matrix $A(w)$ defined in Proposition 4 below satisfies the Hawkins–Simon condition, so that $(I - A(w))^{-1} = \sum_{k \geq 0} A(w)^k$ exists and is nonnegative.

The following proposition states the version of the non-substitution theorem that I use. It delivers, for every wage vector, demand-invariant prices of goods and a linear map from any final consumption vector to the labor demanded (and hence to the factor payments generated) in each country.

Proposition 4 (Factor-content representation). *Suppose Assumption 1 holds. Then for every strictly positive wage vector $w = (w_i)_{i \in \mathcal{I}}$ there exist:*

- a goods-price vector $p(w) \in \mathbb{R}_{++}^{|\mathcal{G}|}$,
- a matrix of intermediate-input coefficients $A(w) \in \mathbb{R}_+^{|\mathcal{G}| \times |\mathcal{G}|}$, whose (j, g) entry is the amount of good j used per unit of good g by the cost-minimizing producing country $i(g; w)$,
- a matrix of unit labor requirements $B(w) \in \mathbb{R}_+^{|\mathcal{I}| \times |\mathcal{G}|}$, whose (i, g) entry equals country i 's labor input per unit of good g if $i = i(g; w)$ and zero otherwise,

such that:

1. For each good g , the cost-minimizing plan to produce one unit of g at prices $(w, p(w))$ is operated by country $i(g; w)$ and uses $B_{i(g; w), g}(w)$ units of country- $i(g; w)$ labor, and $A_{jg}(w)$ units of good j as intermediate for each $j \in \mathcal{G}$. These inputs depend only on w , not on the composition of demand.
2. For any final-consumption vector $c \in \mathbb{R}_+^{|\mathcal{G}|}$ there exists a competitive production plan that delivers c as net output, uses gross output

$$y(w, c) = (I - A(w))^{-1} c,$$

and uses total labor in country i

$$\ell_i(w, c) = \sum_{g \in \mathcal{G}} B_{ig}(w) y_g(w, c).$$

⁶This is a genericity (no-tie) condition: it rules out knife-edge wage vectors at which two countries tie as the least-cost producer of some good. At such ties, the matrices $A(w)$ and $B(w)$ —and hence the factor-content map of Proposition 4—would depend on how production is split among the tied countries. The analysis can be read as applying to all wage vectors off this knife-edge set.

3. Defining $L(w) \equiv B(w)(I - A(w))^{-1}$, total labor demand can be written as

$$\ell(w, c) = L(w) c \in \mathbb{R}_+^{|\mathcal{I}|}.$$

The associated vector of factor payments is

$$E(c; w) \equiv \text{diag}(w) L(w) c,$$

which is linear in c .

4. Prices satisfy the zero-profit factor-content identity

$$p(w) = L(w)^\top w,$$

so that, for any consumption vector c , total expenditure equals total embodied factor payments: $p(w) \cdot c = \sum_{i \in \mathcal{I}} E_i(c; w)$.

Proof. Fix a strictly positive wage vector w . For any candidate goods-price vector p , the unit-cost problems in Assumption 1 define unit-cost functions $c_{ig}(w_i, p)$ for all (i, g) . Under constant returns and competition, an equilibrium goods-price vector must satisfy the zero-profit pricing equations

$$p_g = \min_i c_{ig}(w_i, p) \quad \text{for each } g \in \mathcal{G},$$

and Assumption 1(5) guarantees that a unique strictly positive solution, denoted $p(w)$, exists.

At prices $(w, p(w))$, the cost-minimizing producer of each good g is the unique country $i(g; w)$ from Assumption 1(6), and it operates the unique cost-minimizing input bundle from Assumption 1(4). Let $A_{jg}(w)$ denote the amount of good j used as intermediate per unit of good g by country $i(g; w)$, and let $B_{ig}(w)$ denote country i 's unit labor requirement for good g , equal to the labor input of the cost-minimizing bundle if $i = i(g; w)$ and zero otherwise—countries other than the least-cost producer are inactive in sector g under competition. These inputs solve cost-minimization problems whose data consist solely of $(w, p(w))$, so they are independent of demand, thereby proving part 1.

Let y be the vector of gross outputs across goods, with each good produced by its least-cost producer. For a given final-consumption vector c , the input–output relations imply

$$y = c + A(w) y,$$

and by the Hawkins–Simon condition in Assumption 1(6), $(I - A(w))^{-1}$ exists and is

nonnegative, so that

$$y(w, c) = (I - A(w))^{-1} c \in \mathbb{R}_+^{|\mathcal{G}|}.$$

Total labor used in country i is

$$\ell_i(w, c) = \sum_{g \in \mathcal{G}} B_{ig}(w) y_g(w, c),$$

or in matrix form

$$\ell(w, c) = B(w) y(w, c) = B(w) (I - A(w))^{-1} c.$$

Defining $L(w) \equiv B(w) (I - A(w))^{-1}$ yields $\ell(w, c) = L(w) c$. Factor payments are $w_i \ell_i(w, c)$, so

$$E(c; w) = \text{diag}(w) L(w) c,$$

which is linear in c by construction, proving parts 2 and 3.

For part 4, evaluate the unit cost of each good g at the minimizing bundle of its producer $i(g; w)$:

$$p_g = c_{i(g; w), g}(w_{i(g; w)}, p(w)) = \sum_{i \in \mathcal{I}} w_i B_{ig}(w) + \sum_{j \in \mathcal{G}} p_j(w) A_{jg}(w),$$

where the first sum has a single nonzero term by construction of $B(w)$. In matrix form, $p(w) = B(w)^\top w + A(w)^\top p(w)$, so

$$p(w) = (I - A(w)^\top)^{-1} B(w)^\top w = \left(B(w) (I - A(w))^{-1} \right)^\top w = L(w)^\top w.$$

Consequently, $p(w) \cdot c = w^\top L(w) c = \sum_i w_i \ell_i(w, c) = \sum_i E_i(c; w)$ for any c . $\square$

Proposition 4 formalizes the sense in which the economy's supply side is invariant to the boycott. First, goods prices $p(w)$ are pinned down by wages alone, independent of demand, as are each sector's cost-minimizing unit labor requirement and intermediate input coefficients. Note that firms may use all goods as intermediate inputs and may substitute freely among them; what the non-substitution theorem delivers is that the inputs $A(w)$ and $B(w)$ chosen in equilibrium do not vary with the composition of final demand (see [Acemoglu and Azar, 2020](#), for a modern treatment). Second, inverting the input-output structure yields the total, direct plus indirect, labor content of any final consumption vector: $L(w) = B(w) (I - A(w))^{-1}$ is the matrix of total unit labor requirements, and $E(c; w) = \text{diag}(w) L(w) c$ is the vector of country-by-country factor payments generated in producing c . The linearity of $E(\cdot; w)$ in c is what allows me to express boycotters' equilibrium behavior in terms of net spending budgets on each

country's factor services, as the next proposition shows.

Proposition 5 (Boycotting equilibrium as a net factor-spending problem). *Suppose Assumption 1 holds, so that Proposition 4 applies and defines the factor-content map $E(c; w) = \text{diag}(w)L(w)c$.*

Consider a boycotting equilibrium with wage vector w , goods prices $p(w)$, and allocations $\{c(h), \ell_{i(h)}(h)\}_h$. Let i^ be the boycotting country and B_{i^*} the set of boycotting households in that country. Then for each boycotting household $h \in B_{i^*}$ there exists a vector of “net factor-spending budgets” $b(h) = (b_i(h))_{i \in \mathcal{I}}$ such that the equilibrium allocation $(c(h), \ell_{i^*}(h))$ solves*

$$(c(h), \ell_{i^*}(h)) \in \arg \max_{c, \ell_{i^*}} u(c, \ell_{i^*}; h)$$

subject to

$$\begin{aligned} E_i(c; w) &= b_i(h) \quad \text{for all } i \neq i^*, \\ E_{i^*}(c; w) &= b_{i^*}(h) + w_{i^*} \ell_{i^*}. \end{aligned}$$

In particular, one can take

$$b_i(h) = E_i(c(h); w) \quad \text{for all } i \neq i^*, \quad b_{i^*}(h) = E_{i^*}(c(h); w) - w_{i^*} \ell_{i^*}(h).$$

Proof. Fix a boycotting equilibrium and wages w . Proposition 4 provides the linear factor-content map

$$E(c; w) = \text{diag}(w)L(w)c.$$

Fix a boycotting household $h \in B_{i^*}$. Define

$$b_i(h) \equiv E_i(c(h); w) \quad \text{for all } i \neq i^*, \quad b_{i^*}(h) \equiv E_{i^*}(c(h); w) - w_{i^*} \ell_{i^*}(h).$$

Then the equilibrium allocation $(c(h), \ell_{i^*}(h))$ satisfies

$$E_i(c(h); w) = b_i(h) \quad \text{for all } i \neq i^*, \quad E_{i^*}(c(h); w) = b_{i^*}(h) + w_{i^*} \ell_{i^*}(h),$$

so it is feasible for the optimization problem in the statement.

Suppose, for a contradiction, that $(c(h), \ell_{i^*}(h))$ does not solve this problem. Then there exists $(\tilde{c}, \tilde{\ell}_{i^*})$ such that:

$$\begin{aligned} E_i(\tilde{c}; w) &= b_i(h) \quad \text{for all } i \neq i^*, \\ E_{i^*}(\tilde{c}; w) &= b_{i^*}(h) + w_{i^*} \tilde{\ell}_{i^*}, \end{aligned}$$

so that

$$E_i(\tilde{c}; w) - \mathbf{1}_{\{i=i^*\}} w_{i^*} \tilde{\ell}_{i^*} = E_i(c(h); w) - \mathbf{1}_{\{i=i^*\}} w_{i^*} \ell_{i^*}(h) \quad \text{for all } i \in \mathcal{I},$$

and

$$u(\tilde{c}, \tilde{\ell}_{i^*}; h) > u(c(h), \ell_{i^*}(h); h).$$

Construct a new allocation as follows. Keep wages w and goods prices $p(w)$ unchanged. Keep the allocations of all non-boycotting households and all other boycotters $h' \neq h$ unchanged. Assign household h the new allocation $(\tilde{c}, \tilde{\ell}_{i^*})$.

Let c^{tot} and $\tilde{c}^{\text{tot}}$ denote the aggregate consumption vectors before and after this change. By the linearity of the factor-content map, $E(\tilde{c}^{\text{tot}}; w) - E(c^{\text{tot}}; w) = E(\tilde{c}; w) - E(c(h); w)$. Since the net factor spending of h is unchanged in each country, the labor demanded in each country $i \neq i^*$ is unchanged, and the change in the labor demanded in country i^* ,

$$\frac{E_{i^*}(\tilde{c}^{\text{tot}}; w) - E_{i^*}(c^{\text{tot}}; w)}{w_{i^*}} = \tilde{\ell}_{i^*} - \ell_{i^*}(h),$$

exactly equals the change in household h 's labor supply. Thus, all labor markets continue to clear at wages w . Moreover, by part 4 of Proposition 4, total expenditure equals total embodied factor payments, $p(w) \cdot c = \sum_{i \in \mathcal{I}} E_i(c; w)$ for any bundle c . In particular, $\sum_{i \in \mathcal{I}} b_i(h) = p(w) \cdot c(h) - w_{i^*} \ell_{i^*}(h) = 0$ by household h 's budget constraint (25) in the original equilibrium, so the budget constraint also holds at the new allocation: $p(w) \cdot \tilde{c} = \sum_{i \in \mathcal{I}} b_i(h) + w_{i^*} \tilde{\ell}_{i^*} = w_{i^*} \tilde{\ell}_{i^*}$.

By Proposition 4, there exists a cost-minimizing production plan at wages w that delivers $\tilde{c}^{\text{tot}}$ as net output using the inputs $A(w)$ and $B(w)$. Thus, we can obtain a new competitive equilibrium in which: (i) all non-boycotters' allocations and utilities are unchanged, (ii) all other boycotters $h' \neq h$ are unchanged, (iii) household h attains utility $u(\tilde{c}, \tilde{\ell}_{i^*}; h) > u(c(h), \ell_{i^*}(h); h)$.

This contradicts the definition of a boycotting equilibrium as a competitive equilibrium for which there is no other competitive equilibrium with the same utilities for non-boycotters and a Pareto improvement for the boycotting coalition. Hence no such pair $(\tilde{c}, \tilde{\ell}_{i^*})$ can exist, and $(c(h), \ell_{i^*}(h))$ solves the stated optimization problem. $\square$

I now strengthen Proposition 5 to a full equivalence: the net factor-spending formulation not only characterizes boycotting equilibria but also identifies them. The main additional element required in the converse direction is that non-boycotter utilities pin down wages.

Assumption 2 (Non-boycotter utilities pin down wages). *For any wage vector $w = (w_i)_{i \in \mathcal{I}}$, let $p(w)$ and $L(w)$ be the associated goods-price vector and consumption-to-labor-demand matrix given by Assumption 1 and Proposition 4. Consider the non-*

boycotters' optimal choices at goods prices $p(w)$ and wages w , and let $U^{\text{NB}}(w)$ denote the implied vector of utilities for all non-boycotters.

Assume:

1. For each w , the non-boycotter equilibrium allocation and utility vector $U^{\text{NB}}(w)$ are uniquely determined.
2. The mapping $w \mapsto U^{\text{NB}}(w)$ is injective up to a common normalization of wages: if $U^{\text{NB}}(w') = U^{\text{NB}}(w)$, then $w' = \kappa w$ for some scalar $\kappa > 0$.⁷

Assumption 2 is the central ingredient of the “if” direction of the equivalence, and it is worth explaining why. The net factor-spending problem in Proposition 5 characterizes boycotters' behavior *at given wages*. To establish the converse—that behavior of this form identifies a boycotting equilibrium—one must rule out the possibility that the coalition could make its members better off through an alternative profile that leaves all non-boycotters exactly as well off. Such an alternative profile could, in principle, feature different wages and prices. Assumption 2 closes this door: if an alternative profile delivers the same utilities to all non-boycotters, then, up to the wage normalization, it must feature the same wage vector and, by the demand invariance and uniqueness of zero-profit prices (Assumption 1), the same goods prices. All remaining differences between the two profiles must then lie in the boycotters' allocations, which the net factor-spending budgets discipline.

The resulting equivalence is as follows: a profile of prices, wages, and quantities is a boycotting equilibrium if and only if (i) it satisfies the competitive equilibrium conditions for firms, regular households, and boycotters' budget balance; (ii) each boycotting household maximizes her utility subject to net factor-spending budgets, exactly as in Proposition 5; and (iii) the profile of budgets across coalition members is efficient for the coalition, in the sense that no reallocation of budgets with the same coalition aggregates—each member still spending only her own wage income—makes every boycotter weakly better off and at least one boycotter strictly better off. When the coalition has a representative member, as in the DFS model of Section 2, this budget-efficiency requirement is vacuous. The formal statement, Proposition 6, and its proof are in Appendix A.5, along with an auxiliary assumption that rules out lump-sum transfers across households (Assumption 3).

The following remark shows that Assumption 2, which drives the “if” part of the equivalence, is mild: it suffices that a single non-boycotting household have an indirect utility that is injective in (normalized) wages.

⁷Under constant returns, the zero-profit prices are homogeneous of degree one in wages, $p(\kappa w) = \kappa p(w)$, so households' budget sets—and hence utilities—are invariant to a common rescaling of wages. Injectivity can therefore hold only modulo this normalization, just as the analysis of Section 2 normalizes $w_F = 1$. In what follows, I fix a numéraire by normalizing the wage in a reference country, so that statements such as “ $w' = w$ ” are understood to hold for normalized wage vectors.

Remark 1 (On Assumption 2). *Assumption 2 is implied by the requirement that there exists at least one non-boycotting household h^* whose indirect utility function $V_{h^*}(w)$ is injective in the wage vector w , on wage vectors normalized by the numéraire. In other words, it is enough that for some non-boycotter h^* , on normalized wage vectors, we have*

$$w' \neq w \Rightarrow V_{h^*}(w') \neq V_{h^*}(w),$$

since in that case the full profile $U^{\text{NB}}(w) = (V_h(w))_{h \notin B}$ is automatically injective.

In the environments considered in this paper, non-boycotting households have separable preferences of the form

$$u_h(c, \ell_{i(h)}) = U_h(c) - v_h(\ell_{i(h)}),$$

where $U_h(\cdot)$ is strictly concave, homogeneous of degree one, and smooth in consumption, and $v_h(\cdot)$ is strictly convex with standard Inada conditions in labor. Under these primitives, the indirect utility of at least one non-boycotter (for instance a representative Foreign household) is strictly increasing in its real wage index and hence injective in w up to the normalization. Therefore, Assumption 2 is satisfied in the environments we consider.

Assumption 2 can be further unpacked in terms of real wages, as the following remark and its three examples illustrate.

Remark 2 (Real wages and examples for Assumption 2). *Assumption 2 is most naturally understood in terms of real wages for some “anchor” non-boycotting household h^* . Let $i(h^*)$ denote the country of h^* , let $p(w)$ be the equilibrium goods-price vector under wages w , and let $P(w)$ be the corresponding consumption price index relevant for h^* (Cobb–Douglas, CES, etc., depending on preferences). Define the real wage index of h^* as*

$$R_{h^*}(w) \equiv \frac{w_{i(h^*)}}{P(w)}.$$

Note that $R_{h^}(w)$ is homogeneous of degree zero in w , so its injectivity is understood on normalized wage vectors. If the map $w \mapsto R_{h^*}(w)$ is injective on normalized wage vectors and the indirect utility $V_{h^*}(w)$ of h^* is strictly increasing in $R_{h^*}(w)$, then $V_{h^*}(w)$ is injective in normalized w . In that case Assumption 2 is satisfied because $V_{h^*}(w)$ alone pins down w up to the normalization, and therefore the full vector $U^{\text{NB}}(w)$ is injective up to the normalization.*

This condition is met in many standard environments. The following examples illustrate three familiar classes of preferences for the anchor household h^ under which $V_{h^*}(w)$ is strictly increasing in the real wage index $R_{h^*}(w)$.*

Example 1: (Cobb–Douglas in goods, power disutility of labor). Suppose

$$u(c, \ell) = \sum_{g \in \mathcal{G}} \beta_g \log c_g - \frac{\kappa}{1 + \eta} \ell^{1 + \eta}, \quad \beta_g > 0, \quad \sum_g \beta_g = 1, \quad \kappa > 0, \quad \eta > 0.$$

Let $P(w)$ be the Cobb–Douglas price index associated with $p(w)$. Given $(p(w), w_{i(h^)})$, the household spends all income $y(w) = w_{i(h^*)} \ell(w)$ on consumption, with optimal consumption proportional to $y(w)/P(w)$. The optimal labor supply $\ell(w)$ solves a strictly concave problem in the real wage $R_{h^*}(w) = w_{i(h^*)}/P(w)$, and the resulting indirect utility $V_{h^*}(w)$ is strictly increasing in $R_{h^*}(w)$. If $w \mapsto R_{h^*}(w)$ is injective, then $V_{h^*}(w)$ is injective in w .*

Example 2: (CES in goods, power disutility of labor). Suppose

$$u(c, \ell) = \left(\sum_{g \in \mathcal{G}} \alpha_g c_g^\rho \right)^{1/\rho} - \frac{\kappa}{1 + \eta} \ell^{1 + \eta}, \quad \rho < 1, \quad \alpha_g > 0, \quad \kappa > 0, \quad \eta > 0.$$

Let $P(w)$ be the associated CES price index. The optimal consumption bundle is homothetic, proportional to $y(w)/P(w)$, where $y(w) = w_{i(h^)} \ell(w)$. The labor choice again solves a strictly concave problem in the real wage index $R_{h^*}(w)$, and the indirect utility is strictly increasing in $R_{h^*}(w)$. An injective mapping $w \mapsto R_{h^*}(w)$ therefore implies injectivity of $V_{h^*}(w)$.*

Example 3: (Quasi-linear in income, convex disutility of labor). Suppose

$$u(c, \ell) = \sum_{g \in \mathcal{G}} \beta_g c_g - v(\ell), \quad \beta_g > 0, \quad v'(\cdot) > 0, \quad v''(\cdot) > 0.$$

The household always spends all income $y(w) = w_{i(h^)} \ell(w)$ and chooses c to maximize $\sum_g \beta_g c_g$ subject to $p(w) \cdot c \leq y(w)$, so that maximal utility from consumption is proportional to $y(w)/P(w)$ for an appropriate price index $P(w)$. The labor choice solves a strictly concave problem $\max_\ell \{(w_{i(h^*)}/P(w)) \ell - v(\ell)\}$, implying that indirect utility is strictly increasing in the real wage index $R_{h^*}(w) = w_{i(h^*)}/P(w)$. As above, injectivity of $w \mapsto R_{h^*}(w)$ then yields injectivity of $V_{h^*}(w)$.*

In all three examples, once general equilibrium determines that the real wage index $R_{h^}(w)$ is one-to-one in normalized w (as in the familiar two-country trade models, where, after the normalization $w_F = 1$, a scalar relative wage is pinned down), the anchor household's indirect utility $V_{h^*}(w)$ is injective in w up to the normalization. This ensures that Assumption 2 holds.*

3.2 Discussion

I close this section by discussing extensions of the analysis. Although they build on the DFS model of Section 2, I present them here because they depart from the standard DFS specification while remaining tied to the general environment of Section 3—CES preferences as a special case of it, and strategic interaction between two boycotting coalitions as an extension built on it—and because they exploit the budget characterization of boycotter behavior delivered by the equivalence results above.

Substitution Elasticities. Our baseline model is one in which the elasticity of substitution equals 1. Here, I consider different possible elasticities of substitution that can be captured in my framework by CES preferences:

$$U(\mathbf{q}) = \left(\int_0^1 s(j)^{\frac{1}{\gamma}} q^i(j)^{\frac{\gamma-1}{\gamma}} dj \right)^{\frac{\gamma}{\gamma-1}} \quad (26)$$

In this setting, whenever preferences are not Cobb–Douglas ($\gamma \neq 1$), the expenditure share on each good j depends not only on the exogenous $s(j)$ but also on the endogenous price of good j and the price index P faced by regular households and boycotters. In Appendix A.7, I derive the corresponding balanced trade condition, which has the same form as in the Cobb–Douglas case with endogenous expenditure shares, and I explain how the cutoff characterization of the optimal boycott set in Proposition 2 extends to these preferences.

Comparative Statics. The marginal change in the boycotters’ welfare for a given decrease in Foreign’s welfare captures the slope of the Pareto frontier. An increase in the boycott cutoff j_B yields a marginal decrease in Foreign’s welfare. Home boycotters lose from consuming more expensive domestic goods but gain from higher relative wages. The loss from consuming more expensive domestic products is proportional to the elasticity of the expenditure share on boycotted goods that is shifted to domestic goods to the relative wage $\varepsilon_{S(j_B),\omega}$ (see Appendix A.8). Targeted boycotts are, therefore, more effective when this elasticity is smaller.

Strategic Retaliation. Retaliation occurs when a coalition of foreign households boycotts the home country’s exports in return. This situation leads to strategic considerations in which the share of boycotters endogenously depends on the foreign country’s response. I model this in a setting where $j_B = 1$ and a share λ^F of Foreign boycotts all Home goods (the equivalent for Home boycotters is λ^H). The balanced trade condition

then becomes

$$\omega = \frac{1 - \lambda^F}{1 - \lambda^H} \cdot \frac{S(\bar{j})}{1 - S(\bar{j})} \cdot \frac{l_F}{l_H} \quad (27)$$

We see that if each country values the terms-of-trade gains accruing to its nonboycotters and chooses its share of boycotters accordingly, then a prisoner's dilemma, reminiscent of a trade war, emerges: an increase in the share of boycotters tilts the terms of trade in one's favor.⁸

For instance, consider the parameterization of productivities with $z_F(j) = 1$ for all j and $z_H(j) = \left(\frac{T_H}{T_F}\right)^{\frac{1}{\mu}} \left(\frac{j}{1-j}\right)^{-\frac{1}{\mu}}$ (the two-country DFS representation of Fréchet productivity draws as in [Eaton and Kortum, 2002](#)) with $T_H = T_F = 1$ and $\mu = 1$. In this case, the welfare of a representative Foreign nonboycotter is $\ln\left(\frac{1}{\omega} + 1\right)$. If the Foreign government cares only about the utility of its representative nonboycotter, the best response to a boycott λ^H is to increase λ^F arbitrarily close to 1. However, if both governments are utilitarian, the only Nash equilibrium is one where no country boycotts $\lambda^H = \lambda^F = 0$, as the welfare loss from boycotters going into autarky outweighs the terms-of-trade gains for nonboycotters. Therefore, it is difficult to justify boycotts on the basis of purely internal equity arguments.

4 Conclusion

This paper provides the first systematic analysis of international boycotts within a neo-classical trade framework. Boycotting households maximize their own welfare subject to holding Foreign welfare at a tolerated level. My general equivalence result shows that, in equilibrium, this behavior reduces to maximizing utility subject to fixed expenditure budgets on each country's goods (and equivalently, net spending budgets on each country's factor services), providing conceptual clarity on the economics underlying boycott actions. Optimal boycotts prohibit certain Foreign goods, and boycotters only import goods for which Foreign's comparative advantage is sufficiently large. In addition, when income effects outweigh substitution effects, boycotters optimally work more to increase the budget they spend on domestically produced goods.

I show the equivalence between consumer boycotts and certain forms of government-imposed sanctions or voluntary export restrictions (VERs) imposed by foreign firms, and that the optimal boycott mirrors the uniform structure of optimal import tariffs. However, the effectiveness of boycotts is moderated by leakage effects from non-boycotting households and the risk of strategic retaliation from Foreign. Thus, while boycotts share important economic parallels with traditional trade policy tools, they

⁸[Opp \(2010\)](#) studies tariff wars in a similar Ricardian model with a continuum of goods.

also introduce unique complexities and can play a distinctive role as an instrument of geoeconomic influence.

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A Appendix

A.1 Proof of Equivalence Proposition 1

Starting from a competitive equilibrium (CE) with a boycotter coalition, consider the problem of Foreign households. This problem is standard. A key object is the relative wage of Home vs. Foreign workers $\omega = w_H/w_F$. As in the standard [Dornbusch et al. \(1977\)](#) model, the patterns of trade of regular households are such that there exists an indifference good $\bar{j} = Z^{-1}(\omega)$, such that Foreign imports goods with an index below this cutoff and exports goods with an index above it. Foreign households spend constant Cobb–Douglas shares of their income on each good: $p(j)q^F(j) = s(j)w_F l_F$. Throughout this proof, I assume an interior trade equilibrium, $\bar{j} \in (0, 1)$, and that Z is continuously differentiable and strictly decreasing, so that $\bar{j} = Z^{-1}(\omega)$ is well defined.

Using the demand functions and equilibrium prices, we can integrate to find the total utility for the representative household in the foreign country:

$$U^F = \int_0^1 s(j) \log \left(\frac{s(j)w_F l_F}{p(j)} \right) dj \quad (28)$$

where the price function is given by

$$p(j) = \begin{cases} \frac{w_H}{z_H(j)} & \text{if } j \leq \bar{j} \\ \frac{w_F}{z_F(j)} & \text{if } j > \bar{j} \end{cases} \quad (29)$$

i.e., the Foreign real wage is

$$\frac{w_F}{p(j)} = \begin{cases} \frac{z_H(j)}{\omega} & \text{if } j \leq \bar{j} \\ z_F(j) & \text{if } j > \bar{j} \end{cases} \quad (30)$$

and welfare is

$$U^F = \int_0^{\bar{j}} s(j) \log \left(\frac{z_H(j)}{\omega} \right) dj + \int_{\bar{j}}^1 s(j) \log (z_F(j)) dj + \int_0^1 s(j) \log (s(j)l_F) dj \quad (31)$$

Observe here that Foreign's welfare depends only on the boycotting coalition's choices insofar as they affect the relative wage ω and the trade cutoff $\bar{j}$. Differentiating (31) with respect to ω , and noting that the terms involving the endogenous cutoff $\bar{j}(\omega)$ vanish by an envelope argument—at the cutoff, the marginal good is priced identically in the two countries, so $s(\bar{j}) \log (z_H(\bar{j})/(\omega z_F(\bar{j}))) \bar{j}'(\omega) = s(\bar{j}) \log (Z(\bar{j})/\omega) \bar{j}'(\omega) = 0$

since $Z(\bar{j}) = \omega$ —we obtain

$$\frac{dU^F}{d\omega} = -S(\bar{j})\frac{1}{\omega} < 0. \quad (32)$$

Foreign utility is strictly decreasing in the relative wage of Home. So there exists a unique mapping Ω , defined on the attainable range of U^F , such that $U^F = v^F \iff \omega = \Omega(v^F)$.

From trade balance, Foreign income equals world spending on Foreign-produced goods:

$$w_F l_F = (1 - \lambda)(1 - S(\bar{j}))w_H l_H + \lambda \mathcal{E}_F + (1 - S(\bar{j}))w_F l_F \quad (33)$$

where $\mathcal{E}_F$ denotes the expenditure of each boycotter on Foreign goods.

Let's normalize $w_F = 1$ without loss of generality, so that Foreign prices $p_F(j) = 1/z_F(j)$ do not vary with ω and $\mathcal{E}_F = \int_0^1 p_F(j) q_F^B(j) dj$ depends only on the boycotters' demand schedule. The budget of domestic boycotters on Foreign goods implied by trade balance is

$$\mathcal{E}_F = \frac{1}{\lambda} (S(\bar{j})l_F - (1 - \lambda)(1 - S(\bar{j}))\omega l_H) \quad (34)$$

Differentiating (34) with respect to ω , and accounting for the endogenous terms through the cutoff $\bar{j}(\omega) = Z^{-1}(\omega)$ —which, in contrast to the envelope argument for U^F above, do not vanish here but reinforce the direct effect—we obtain

$$\frac{d\mathcal{E}_F}{d\omega} = \frac{1}{\lambda} \left(\underbrace{s(\bar{j}) \bar{j}'(\omega) [l_F + (1 - \lambda)\omega l_H]}_{<0} - \underbrace{(1 - \lambda)(1 - S(\bar{j}))l_H}_{\geq 0} \right) < 0, \quad (35)$$

where $\bar{j}'(\omega) = 1/Z'(\bar{j}) < 0$ because Z is strictly decreasing. Boycotters' budget on Foreign goods is strictly decreasing in Home's relative wage. So the trade-balance locus (34) defines a strictly decreasing—hence invertible—mapping $\omega \mapsto b_F(\omega)$ such that $\mathcal{E}_F = b_F(\omega)$ and, since the boycotters' budget constraint binds, $\mathcal{E}_H = \omega l_H - b_F(\omega)$. With a slight abuse of notation, let $b_F(v^F) = b_F \circ \Omega(v^F)$. Then we have the chain of equivalences

$$U^F = v^F \iff \omega = \Omega(v^F) \iff \mathcal{E}_F = b_F(v^F) \ \& \ \mathcal{E}_H = \omega l_H - b_F(v^F), \quad (36)$$

where each step follows from the strict monotonicity—hence invertibility—of the corresponding mapping.

I now use (36) to establish the two directions of the proposition.

(Only if). Let the tuple be a CE with a boycotter coalition with Foreign utility v^F , equilibrium relative wage $\omega = \Omega(v^F)$, and define $b_F(v^F)$ as above. Conditions 1, 2, and 4 of Definition 2 coincide with conditions (i), (ii), and (v) of the definition of a CE with a boycotter coalition, so it remains to verify the individual boycotters' optimality condition. In the coalition's problem, the constraint (5) restricts the coalition to demand schedules that induce the relative wage $\Omega(v^F)$; by (36), a demand schedule induces this relative wage if and only if its expenditure split between Foreign and Home goods is $(b_F(v^F), \omega l_H - b_F(v^F))$. Hence, the coalition's constraint set is exactly the set of demand schedules with this fixed expenditure split. Every schedule in this set induces the same market-clearing wages and—by the non-substitution theorem—the same prices, so the coalition's internalization of wages has no further bite within its constraint set: its optimal bundle maximizes utility at the fixed equilibrium prices subject to the two expenditure constraints (12) and (11) with $b_F = b_F(v^F)$. Since boycotters share identical homothetic preferences and identical wage incomes, this is precisely the problem (10) that each atomistic boycotter solves. The tuple is therefore a CE with individual boycotters with fixed budgets $b_F(v^F)$.

(If). Conversely, let the tuple be a CE with individual boycotters with fixed budgets b_F and equilibrium relative wage ω , and define v^F as the Foreign utility at ω , so that $\omega = \Omega(v^F)$ and $b_F = b_F(v^F)$ by (36). Conditions (i), (ii), and (v) of the definition of a CE with a boycotter coalition hold immediately, and the Foreign utility constraint (5) holds by construction of v^F . For the coalition's optimality and wage consistency, conditions (iii) and (iv): any coalition demand schedule satisfying (5) must, by (36), have the expenditure split $(b_F(v^F), \omega l_H - b_F(v^F))$ and therefore induces the same market-clearing wages $\{w_H, w_F\}$ —so condition (iv) holds—and the same prices. Among all such schedules, the bundle chosen by the individual boycotters attains the maximal utility, because it solves the same constrained utility-maximization problem at the same prices. Hence, condition (iii) holds, and the tuple is a CE with a boycotter coalition with Foreign utility v^F . This completes the proof.

A.2 Proof of Existence of Boycotted Goods Lemma 1

At prices

$$p_H(j) = \frac{w_H}{z_H(j)} \quad \& \quad p_F(j) = \frac{w_F}{z_F(j)}, \quad (37)$$

the individual boycotter's problem (10) has two expenditure constraints, (11) and (12), with associated multipliers μ_F and μ_H . For interior $q_H^B(j)$ and $q_F^B(j)$ choices, the first-

order conditions for domestic boycotters are:

$$\frac{s(j)}{q_H^B(j) + q_F^B(j)} = \mu_H p_H(j) \quad \forall j \in [0, 1] \quad \& \quad q_H^B(j) > 0 \quad (38)$$

$$\frac{s(j)}{q_H^B(j) + q_F^B(j)} = \mu_F p_F(j) \quad \forall j \in [0, 1] \quad \& \quad q_F^B(j) > 0 \quad (39)$$

Suppose by contradiction that (i) $\forall j \in [0, 1] \quad q_F^B(j) > 0$. Suppose (ii) that there exists a set of positive measure Y such that $q_H^B(\cdot) > 0$ on Y . On Y , (38) and (39) hold jointly, so that $p_H(j)/p_F(j) = \mu_F/\mu_H$ on Y . Since $p_H(j)/p_F(j) = \omega/Z(j)$ is strictly increasing in j , it equals the constant μ_F/μ_H at most at one point, so (38) and (39) cannot hold jointly on Y . That is, given (i), assumption (ii) is a contradiction. Therefore, on any set of positive measure, $q_H^B(\cdot) = 0$, and so the expenditure on domestic goods is zero. This contradicts the lemma's assumption that boycotters' expenditure on Home goods is positive, $w_H l_H - b_F > 0$. Thus, (i) is false. Therefore, if $b_F < w_H l_H$, the set of completely boycotted Foreign goods $\mathcal{J}_B = \{j \in [0, 1]; q_F^B(j) = 0\}$ is nonempty.

The same argument shows that the Home-sourced and Foreign-sourced sets are essentially disjoint, and the first-order conditions determine the sourcing pattern: each good is sourced from the country with the lower shadow cost, $\mu_H p_H(j)$ versus $\mu_F p_F(j)$. Since $p_H(j)/p_F(j) = \omega/Z(j)$ is strictly increasing, the Home-sourced set is $\mathcal{J}_B = \{j : \omega/Z(j) \leq \mu_F/\mu_H\} = [0, j_B]$ (almost everywhere), an interval with cutoff j_B given by $\omega/Z(j_B) = \mu_F/\mu_H$ when interior. The multipliers follow from the expenditure constraints. Summing (38) over the Home-sourced set and (39) over its complement yields

$$p_H(j)q_H^B(j) = s(j) \frac{w_H l_H - b_F}{S(j_B)} \quad \forall j \leq j_B, \quad \mu_H = \frac{S(j_B)}{w_H l_H - b_F}, \quad (40)$$

$$p_F(j)q_F^B(j) = s(j) \frac{b_F}{1 - S(j_B)} \quad \forall j > j_B, \quad \mu_F = \frac{1 - S(j_B)}{b_F}, \quad (41)$$

where $S(j_B) = \int_0^{j_B} s(j) dj$ is the preference mass of the Home-sourced set. That is, conditional on each sourcing set, Cobb–Douglas expenditure shares renormalize by the set's preference mass, and the boycotters' total spending on Home and Foreign goods equals their budgets, $w_H l_H - b_F$ and b_F , by construction. When the boycott constraint does not bind, $\mu_H = \mu_F = 1/(w_H l_H)$, the budget split is proportional to the preference masses, $b_F = (1 - S(j_B)) w_H l_H$, the sourcing cutoff coincides with the trade cutoff ($j_B = \bar{j}$), and the unconditional Cobb–Douglas shares $p_i(j)q_i^B(j) = s(j)w_H l_H$ obtain. A binding boycott corresponds to $\mu_F > \mu_H$ and $j_B > \bar{j}$. Note that we have shown, in passing, that boycotters choose disjoint sets of Home and Foreign goods in which they consume positive amounts of either Home or Foreign goods.

A.3 Proof of Proposition 2

Consider a CE with individual boycotters with fixed budgets $(w_H l_H - b_F, b_F)$, and let $\mathcal{J}_B$ denote the set of Home-sourced (boycotted) goods. Then, domestic income is:

$$w_H l_H = \underbrace{(1 - \lambda) \int_0^{\bar{j}} p(j) q^H(j) dj}_{\text{regular domestic households}} + \underbrace{\lambda \int_{\mathcal{J}_B} p_H(j) q_H^B(j) dj}_{\text{domestic boycotters}} + \underbrace{\int_0^{\bar{j}} p(j) q^F(j) dj}_{\text{Foreign households}} \quad (42)$$

By Appendix A.2, the boycotters' expenditure on the Home-sourced set equals their Home budget, $\int_{\mathcal{J}_B} p_H(j) q_H^B(j) dj = w_H l_H - b_F$, regardless of the composition of $\mathcal{J}_B$. Recalling that $S(\mathbf{q}_H^B) = 1 - b_F / (w_H l_H)$ denotes the boycotters' expenditure share on Home-sourced goods and summing the expenditures as previously, we obtain

$$w_H l_H = (1 - \lambda) S(\bar{j}) w_H l_H + \lambda S(\mathbf{q}_H^B) w_H l_H + S(\bar{j}) w_F l_F \quad (43)$$

Simplifying the equation above yields

$$\omega = \frac{S(\bar{j})}{1 - \lambda S(\mathbf{q}_H^B) - (1 - \lambda) S(\bar{j})} \cdot \frac{l_F}{l_H} \quad (44)$$

Using the demand functions and equilibrium prices, we can integrate to find the total utility for the representative household in the foreign country:

$$U^F = \int_0^1 s(j) \log \left(\frac{s(j) w_F l_F}{p(j)} \right) dj \quad (45)$$

where the price function is given by

$$p(j) = \begin{cases} \frac{w_H}{z_H(j)} & \text{if } j \leq \bar{j} \\ \frac{w_F}{z_F(j)} & \text{if } j > \bar{j} \end{cases} \quad (46)$$

i.e., the Foreign real wage is

$$\frac{w_F}{p(j)} = \begin{cases} \frac{z_H(j)}{\omega} & \text{if } j \leq \bar{j} \\ z_F(j) & \text{if } j > \bar{j} \end{cases} \quad (47)$$

and welfare is

$$U^F = \int_0^{\bar{j}} s(j) \log \left(\frac{z_H(j)}{\omega} \right) dj + \int_{\bar{j}}^1 s(j) \log (z_F(j)) dj + \int_0^1 s(j) \log (s(j) l_F) dj \quad (48)$$

Observe here that Foreign's welfare depends on the boycott only insofar as it affects the relative wage ω and the trade cutoff $\bar{j}$ and that, by (43), these depend on the

boycott only through the budget split $(w_H l_H - b_F, b_F)$. Therefore, from (48), any set of boycotted goods consistent with the same budgets yields the same utility for Foreign households, v^F . Among all such sets, the boycotters' own optimality pins down the composition: by the first-order conditions in Appendix A.2, each good is sourced from the country with the lower shadow cost, and since the domestic price premium $p_H(j)/p_F(j) = \omega/Z(j)$ is strictly increasing in j , the Home-sourced set is an interval $[0, j_B]$ (almost everywhere) with cutoff $\omega/Z(j_B) = \mu_F/\mu_H$. For a boycott, b_F lies weakly below the unconstrained level of spending on Foreign goods, so $\mu_F \geq \mu_H$ and $j_B \geq \bar{j}$: the boycotted set is convex, leaves no set of positive measure between itself and the trade cutoff, and extends from the trade margin into the goods in which Foreign's comparative advantage is smallest. Intuitively, the welfare of Home boycotters depends directly on the composition of $\mathcal{J}_B$ in that boycotted goods with higher index value j are more expensive for Home to produce: replacing a Home-sourced good of high index with one of lower index, holding the budget split fixed, leaves (43), the relative wage, the trade cutoff, and hence Foreign's welfare unchanged, while reducing the cost of the boycotters' consumption basket. Thus, a set $\mathcal{J}_B$ that does not satisfy Proposition 2 cannot be optimal.

A.4 Proof of Optimal Labor Supply Proposition 3

For a given level of labor l_i (and hence income $w_i l_i$), the problem of a household facing arbitrary prices $p(j)$ is standard and is

$$\max_{\{q^i(j)\}} \int_0^1 s(j) \frac{q^i(j)^{1-\gamma} - 1}{1-\gamma} dj \quad \text{s.t.} \quad \int_0^1 p(j) q^i(j) dj = w_i l_i.$$

Introducing a Lagrange multiplier μ for the budget constraint, the Lagrangian is

$$\mathcal{L} = \int_0^1 s(j) \frac{q^i(j)^{1-\gamma} - 1}{1-\gamma} dj - \mu \left[\int_0^1 p(j) q^i(j) dj - w_i l_i \right].$$

The first-order condition with respect to $q^i(j)$ is

$$\frac{\partial \mathcal{L}}{\partial q^i(j)} : \quad s(j) q^i(j)^{-\gamma} - \mu p(j) = 0.$$

Thus,

$$q^i(j)^{-\gamma} = \mu \frac{p(j)}{s(j)} \implies q^i(j) = \left(\frac{s(j)}{\mu p(j)} \right)^{1/\gamma}.$$

To pin down μ , substitute the solution into the budget constraint:

$$\int_0^1 p(j) \left(\frac{s(j)}{\mu p(j)} \right)^{1/\gamma} dj = w_i l_i.$$

This can be rewritten as

$$\left(\frac{1}{\mu} \right)^{1/\gamma} \int_0^1 s(j)^{1/\gamma} p(j)^{1-1/\gamma} dj = w_i l_i.$$

It is convenient to define the auxiliary integral

$$\Phi \equiv \int_0^1 s(j)^{\frac{1}{\gamma}} p(j)^{\frac{\gamma-1}{\gamma}} dj \quad (49)$$

and the associated ideal price index

$$P \equiv \Phi^{\frac{\gamma}{\gamma-1}} = \left(\int_0^1 s(j)^{\frac{1}{\gamma}} p(j)^{\frac{\gamma-1}{\gamma}} dj \right)^{\frac{\gamma}{\gamma-1}}, \quad (50)$$

which is increasing in each price $p(j)$: for $\gamma > 1$, Φ is increasing in $p(j)$ and the outer exponent $\frac{\gamma}{\gamma-1}$ is positive, while for $\gamma < 1$, Φ is decreasing in $p(j)$ and the outer exponent is negative. Then, the above condition reads

$$\left(\frac{1}{\mu} \right)^{1/\gamma} \Phi = w_i l_i,$$

so that

$$\mu = \left(\frac{\Phi}{w_i l_i} \right)^{\gamma}.$$

Plug this back into the optimal demand:

$$q^i(j) = \left(\frac{s(j)}{p(j)} \right)^{1/\gamma} \frac{w_i l_i}{\Phi},$$

Plug the optimal consumption back into the utility from consumption. The consumption part of utility becomes

$$\int_0^1 s(j) \frac{q^i(j)^{1-\gamma} - 1}{1-\gamma} dj.$$

Further simplification of this expression yields

$$\frac{(w_i l_i)^{1-\gamma}}{1-\gamma} \Phi^{\gamma-1} \Phi = \frac{(w_i l_i)^{1-\gamma} \Phi^{\gamma}}{1-\gamma} = \frac{1}{1-\gamma} \left(\frac{w_i l_i}{P} \right)^{1-\gamma},$$

where the last equality uses $\Phi^{\gamma} = P^{\gamma-1}$ from the definition of the ideal price index (50).

Subtracting the disutility of labor, the household's indirect utility is

$$V(w_i, l_i) = \frac{\left(\frac{w_i l_i}{P}\right)^{1-\gamma} - 1}{1-\gamma} - \frac{l_i^{1+\eta}}{1+\eta},$$

which is Equation (16) in the main text: real income $w_i l_i / P$, deflated by the ideal price index, enters the consumption block.

Now, the household chooses l_i to maximize $V(w_i, l_i)$. Differentiate with respect to l_i :

$$\frac{\partial V}{\partial l_i} = \left(\frac{w_i}{P}\right)^{1-\gamma} l_i^{-\gamma} - l_i^\eta = 0.$$

Thus, the first-order condition is

$$\left(\frac{w_i}{P}\right)^{1-\gamma} = l_i^{\eta+\gamma}.$$

It follows that the optimal labor supply is given by

$$l_i^* = \left(\frac{w_i}{P}\right)^{\frac{1-\gamma}{\eta+\gamma}}.$$

Now, consider a domestic boycotter. Because she rejects Foreign varieties on $\mathcal{J}_B$ and buys only domestic varieties at price $p_H(j)$ on that set, her effective price index is

$$P_B = \left(\int_0^1 s(j)^{\frac{1}{\gamma}} p_B(j)^{\frac{\gamma-1}{\gamma}} dj\right)^{\frac{\gamma}{\gamma-1}} \quad (51)$$

where $p_B(j) = p_H(j)$ for boycotted goods $j \leq j_B$ and $p_B(j) = p(j)$, the world price, for non-boycotted goods. Let P_H denote the corresponding ideal price index of a regular Home household, computed from world prices $p(j)$.

Since domestic prices $p_H(j)$ exceed the free-trade prices $p(j)$ on the positive-measure set $[\bar{j}, j_B]$, and since the ideal price index (50) is increasing in each price, we have $P_B > P_H$. Both household types face the same wage w_H , so their optimal labor supplies satisfy

$$\frac{l^{B*}}{l_H^*} = \left(\frac{P_H}{P_B}\right)^{\frac{1-\gamma}{\eta+\gamma}}.$$

Because $P_H/P_B < 1$ and $\eta + \gamma > 0$, this ratio exceeds one if and only if $1 - \gamma < 0$.

Therefore, $l^{B*} > l_H^*$ if and only if $\gamma > 1$; in the log case, the two labor supplies coincide, and for $\gamma < 1$ boycotters work less than regular Home households. This proves Proposition 3.

A.5 Statement and Proof of the General Equivalence Proposition 6

This appendix states and proves the equivalence discussed in Section 3. In addition to Assumptions 1 and 2, the result uses the following assumption, which formalizes the absence of lump-sum transfers across households.

Assumption 3 (No lump-sum transfers). *Households receive only their own wage income $w_{i(h)}\ell_{i(h)}(h)$; there are no lump-sum transfers across households. Profits are zero under constant returns and perfect competition.*

Proposition 6 (Equivalence: boycotting equilibrium and net factor-spending formulation). *Suppose Assumptions 1, 2, and 3 hold, so that Proposition 4 applies.*

Let i^ be the boycotting country and B_{i^*} the set of boycotting households in that country. A profile*

$$(p, w, \{c(h), \ell_{i(h)}(h)\}_h, \{y_i, \ell_i\}_i)$$

is a boycotting equilibrium in the sense that it satisfies the competitive equilibrium conditions for firms and non-boycotters and implements a point on the Pareto frontier of $\{u(c(h), \ell_{i^}(h); h)\}_{h \in B_{i^*}}$ subject to keeping all non-boycotter utilities fixed, if and only if:*

1. *The profile satisfies the competitive equilibrium conditions for firms and non-boycotters, Equations (22), (23), and (24).*
2. *For the associated wage vector w , the factor-content map $E(\cdot; w)$ is given by $E(c; w) = \text{diag}(w)L(w)c$.*
3. *For each boycotting household $h \in B_{i^*}$ there exists a vector of net factor-spending budgets $b(h) = (b_i(h))_{i \in \mathcal{I}}$ such that the equilibrium allocation $(c(h), \ell_{i^*}(h))$ solves*

$$(c(h), \ell_{i^*}(h)) \in \arg \max_{c, \ell_{i^*}} u(c, \ell_{i^*}; h)$$

subject to

$$E_i(c; w) = b_i(h) \quad \text{for all } i \neq i^*,$$

$$E_{i^*}(c; w) = b_{i^*}(h) + w_{i^*} \ell_{i^*}.$$

4. *The profile of budgets $\{b(h)\}_{h \in B_{i^*}}$ is efficient for the coalition: defining the indirect utility over budgets*

$$V_h(b; w) \equiv \max_{c, \ell_{i^*}} u(c, \ell_{i^*}; h) \quad \text{s.t.} \quad E_i(c; w) = b_i \quad \forall i \neq i^*, \quad E_{i^*}(c; w) = b_{i^*} + w_{i^*} \ell_{i^*},$$

there is no alternative profile of budgets $\{b'(h)\}_{h \in B_{i^*}}$ with the same coalition aggregates, $\sum_{h \in B_{i^*}} b'(h) = \sum_{h \in B_{i^*}} b(h)$, and with $\sum_{i \in \mathcal{I}} b'_i(h) = 0$ for every h , such that $\{V_h(b'(h); w)\}_{h \in B_{i^*}}$ Pareto dominates $\{V_h(b(h); w)\}_{h \in B_{i^*}}$.

In particular, given any boycotting equilibrium, one can take

$$b_i(h) = E_i(c(h); w) \quad \text{for all } i \neq i^*, \quad b_{i^*}(h) = E_{i^*}(c(h); w) - w_{i^*} \ell_{i^*}(h).$$

Condition 4 disciplines the distribution of budgets across coalition members: a reallocation of net factor-spending budgets across boycotters (each member still spending only her own wage income) leaves wages, prices, and all non-boycotter utilities unchanged, so the coalition exhausts its Pareto frontier only if no such reallocation benefits its members. Throughout, I assume that the maxima defining V_h are attained, as they are in our DFS environments where the households' choice set is compact. When the coalition has a representative member, as in the DFS model of Section 2, condition 4 is vacuous, since the aggregate restriction then pins down the budget vector itself; with identical members holding symmetric budgets and concave utility u , it holds automatically because V_h inherits concavity from u over its affine constraint set.

Proof. (Only if). Suppose we have a boycotting equilibrium as defined in the statement. Condition 1 holds by definition, and condition 2 follows from Assumption 1 and Proposition 4. Condition 3, with the budgets $b(h)$ defined in the statement, is precisely the content of Proposition 5, whose proof applies verbatim.

For condition 4, suppose by contradiction that there is a profile of budgets $\{b'(h)\}_{h \in B_{i^*}}$ with the same coalition aggregates, $\sum_h b'(h) = \sum_h b(h)$, with $\sum_i b'_i(h) = 0$ for every h , and whose indirect utilities $\{V_h(b'(h); w)\}_h$ Pareto dominate $\{V_h(b(h); w)\}_h$. Assign to each boycotter h a bundle $(c'(h), \ell'_{i^*}(h))$ attaining $V_h(b'(h); w)$, keeping wages w , goods prices $p(w)$, and all non-boycotters' allocations unchanged. Since $\sum_i b'_i(h) = 0$, each boycotter's budget constraint holds at the new allocation, and by the linearity of the factor-content map, the coalition's total factor content in each country $i \neq i^*$ and its total net factor spending in country i^* are unchanged. Arguing exactly as in the proof of Proposition 5, all goods and labor markets then clear at wages w . The new profile therefore satisfies conditions (i) and (ii) in the definition of a boycotting equilibrium and Pareto improves the coalition, contradicting that the original profile implements a point on the coalition's Pareto frontier.

(If). Conversely, suppose the profile satisfies conditions 1–4. Assume, for a contradiction, that it is not a boycotting equilibrium in the sense of the definition. Then there exists another profile

$$(p', w', \{c'(h), \ell'_{i(h)}(h)\}_h, \{y'_i, \ell'_i\}_i)$$

satisfying the competitive equilibrium conditions for firms and non-boycotters such that:

1. all non-boycotters have the same utilities in the two profiles,
2. all boycotters have weakly higher utility in the alternative profile, and at least one boycotter is strictly better off.

By Assumption 2, equality of the non-boycotter utility profiles across the two profiles implies $w' = \kappa w$ for some scalar $\kappa > 0$; fixing the wage normalization (the same numéraire in both profiles) yields $w' = w$. By the uniqueness of the zero-profit price vector in Assumption 1(5), this also implies $p' = p(w) = p$. Hence, both profiles share the same wage and price vectors. This is where the injectivity of non-boycotter utilities in wages does its work: it anchors the alternative profile to the same price system, so that all remaining differences between the two profiles must lie in the boycotters' allocations.

Given (p, w) and no lump-sum transfers (Assumption 3), each non-boycotter faces the same budget set and solves the same utility-maximization problem in both profiles, whose solution is unique by Assumption 2(1). Therefore, their allocations coincide:

$$c'(h) = c(h), \quad \ell'_{i(h)}(h) = \ell_{i(h)}(h) \quad \text{for all non-boycotters } h.$$

For each boycotter $h \in B_{i^*}$, define the budgets implied by the alternative profile:

$$b'_i(h) \equiv E_i(c'(h); w) \quad \text{for all } i \neq i^*, \quad b'_{i^*}(h) \equiv E_{i^*}(c'(h); w) - w_{i^*} \ell'_{i^*}(h).$$

Since admissible alternative profiles satisfy each boycotter's budget constraint (25), and total expenditure equals total embodied factor payments by part 4 of Proposition 4, $p \cdot c'(h) = \sum_i E_i(c'(h); w) = w_{i^*} \ell'_{i^*}(h)$, so $\sum_i b'_i(h) = 0$ for every h . Moreover, since non-boycotters' allocations and labor supplies are identical in the two profiles, market clearing for goods and labor at the common wage vector w implies that the coalition's aggregate factor content in each country $i \neq i^*$ and its aggregate net factor spending in country i^* coincide across the two profiles:

$$\sum_{h \in B_{i^*}} b'(h) = \sum_{h \in B_{i^*}} b(h).$$

Finally, each boycotter's alternative allocation $(c'(h), \ell'_{i^*}(h))$ is feasible for her own net factor-spending problem with budgets $b'(h)$, so $u(c'(h), \ell'_{i^*}(h); h) \leq V_h(b'(h); w)$, while condition 3 gives $u(c(h), \ell_{i^*}(h); h) = V_h(b(h); w)$. The assumed Pareto domination of the boycotters' utility profile then implies that $\{V_h(b'(h); w)\}_h$ Pareto dominates $\{V_h(b(h); w)\}_h$, with the budget profile $\{b'(h)\}$ satisfying the aggregation and budget-

balance restrictions above. This contradicts condition 4.

Therefore, no such alternative profile can exist. The original profile must be a boycotting equilibrium in the sense of the definition, completing the “if” direction and the proof of the equivalence. $\square$

A.6 Expenditures and Excess Factor Demands in the General Model

The proofs of Propositions 4 and 5 appear in the main text, and Proposition 6 is stated and proven in Appendix A.5. This appendix complements them by deriving the accounting identity that links boycotters’ expenditures on each country’s goods to their excess demands for each country’s labor, which underlies the interpretation of the net factor-spending budgets. Consider a competitive firm in country i producing good g with a constant-returns-to-scale (CRS) production function $F_{ig}(l_{ig}, M_{ig})$. A convenient way to formulate the firm’s choices is to consider the problem:

$$\min_{l_{ig}, M_{ig}} w^i l_{ig} + p_{m,ig} M_{ig} \quad \text{subject to} \quad F_{ig}(l_{ig}, M_{ig}) = 1,$$

which defines the unit cost function

$$c_{ig}(w^i, p_{m,ig}) = \min\{w^i l_{ig} + p_{m,ig} M_{ig} : F_{ig}(l_{ig}, M_{ig}) = 1\}.$$

Let $w \in \mathbb{R}^{|Z|}$ be the vector of wages across countries. By Shephard’s lemma, the partial derivative of the unit cost function with respect to the wage gives the *direct* unit labor requirement of sector (i, g) :

$$\frac{\partial c_{ig}(w^i, p_{m,ig})}{\partial w^i} = B_{ig}(w)$$

in the notation of Proposition 4. Accounting for the labor embodied in intermediate inputs through the input–output structure, the *total* (direct plus indirect) labor content of final consumption is given by the consumption-to-labor-demand matrix

$$\mathcal{L}(\{w^i\}) \equiv L(w) = B(w)(I - A(w))^{-1} \in \mathbb{R}_+^{|Z| \times |\mathcal{G}|},$$

where $[\mathcal{L}(\{w^i\})]_{ig}$ represents the total amount of country- i labor used, directly and through intermediates, per unit of final consumption of good g .

In equilibrium, the price of good g equals its unit cost, and iterating the zero-profit conditions through the input–output structure (part 4 of Proposition 4) expresses the

vector of goods prices $p \in \mathbb{R}^G$ as the total embodied factor payments:

$$p_g = \sum_{i \in \mathcal{I}} w^i [\mathcal{L}(\{w^i\})]_{ig},$$

or in vector notation,

$$p = \mathcal{L}(\{w^i\})^\top w, \quad (52)$$

Define the excess demand (or the factor content of boycotters' consumption) as

$$\bar{L} \equiv l - \mathcal{L}(\{w^i\}) c^{B_i}$$

Multiplying both sides of this equation by the wage vector w (taking the dot product) gives:

$$w \cdot \bar{L} = w \cdot [l - \mathcal{L}(\{w^i\}) c^{B_i}].$$

From the production side (52), we can make expenditures appear by developing the expression and substituting in prices.

$$w \cdot [l - \mathcal{L}(\{w^i\}) c^{B_i}] = w \cdot l - (\mathcal{L}(\{w^i\})^\top w) \cdot c^{B_i} = w \cdot l - p \cdot c^{B_i}.$$

Therefore, we obtain:

$$p \cdot c^{B_i} = w \cdot (l - \bar{L}).$$

This identity shows that, given wages, fixing the boycotters' total expenditure on goods $p \cdot c^{B_i}$ is the same as fixing the excess demand $\bar{L}$ that their consumption leaves available for the rest of the economy, country by country. It is the general-model counterpart of the expenditure-split characterization in Proposition 1 and explains why the budgets $b(h)$ in Propositions 5 and 6 can be read interchangeably as spending on each country's goods or on each country's factor services.

A.7 Substitution Elasticities

For $\gamma \neq 1$, we can define the trade price index and the boycotter price index

$$P(\gamma) = \left(\int_0^1 s(j) p(j)^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}} \quad (53)$$

$$P_B(\gamma) = \left(\int_0^1 s(j) p_B(j)^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}} \quad (54)$$

where $p(j)$ is the world price of good j and $p_B(j)$ is the effective price schedule faced by boycotters: $p_B(j) = p_H(j)$, the domestic producer price, on the boycotted (domestically sourced) set $j \leq j_B$, and $p_B(j) = p(j)$ on non-boycotted goods $j > j_B$. Then, the

expenditure shares on good j for regular households and boycotters are, respectively,

$$\text{share}(j; \gamma) = s(j) \left(\frac{p(j)}{P(\gamma)} \right)^{1-\gamma} \quad (55)$$

$$\text{share}^B(j; \gamma) = s(j) \left(\frac{p_B(j)}{P_B(\gamma)} \right)^{1-\gamma} \quad (56)$$

Aggregating these shares,

$$S(j; \gamma) = \int_0^j \text{share}(j'; \gamma) dj' \quad (57)$$

$$S^B(j; \gamma) = \int_0^{j'} \text{share}^B(j'; \gamma) dj' \quad (58)$$

Domestic income is

$$w_H l_H = (1 - \lambda) S(\bar{j}; \gamma) w_H l_H + \lambda S^B(j_B; \gamma) w_H l_H + S(\bar{j}; \gamma) w_F l_F \quad (59)$$

We obtain the new BT condition:

$$\omega = \frac{S(\bar{j}; \gamma)}{1 - \lambda S^B(j_B; \gamma) - (1 - \lambda) S(\bar{j}; \gamma)} \cdot \frac{l_F}{l_H} \quad (60)$$

We obtain the same balanced trade condition as before, with expenditure shares now endogenous to prices and price indices. The arguments behind Proposition 1, Lemma 1, and Proposition 2 then extend along the same steps as in the Cobb–Douglas case—the strict monotonicity of Foreign welfare and of the boycotters’ Foreign budget in the relative wage, the first-order conditions delivering disjoint sourcing sets, and the cutoff structure of the optimal boycott—with the endogenous shares (55) and (56) in place of $s(j)$; I omit the rederivation.

A.8 Comparative Statics

Throughout this appendix, the comparative statics are taken in the prohibition form of the boycott, as in Section 2.4: the instrument is the boycott cutoff j_B , boycotters buy goods in $[0, j_B]$ domestically, and, with Cobb–Douglas preferences and a single budget, they spend the share $S(j_B) = \int_0^{j_B} s(j) dj$ of their income on them.

Proposition 7. *A reduction in $v^F \downarrow$ comes with an increase in the optimal boycott cutoff ($j_B \uparrow$), an increase in the relative wage of Home workers $\omega \uparrow$, and an increase in the set of Foreign-produced goods $\bar{j} \downarrow$. In particular, the marginal welfare changes are:*

$$\frac{dv^F}{dj_B} = -S(\bar{j}) \frac{d\omega}{\omega dj_B} \leq 0 \quad (61)$$

and the marginal change in Home boycotters' welfare relative to the change in Foreign's welfare is

$$\frac{dU^B}{-dv^F} = \frac{1}{S(\bar{j})} \cdot \left(1 - S(j_B) + S(j_B) \cdot \log \left(\underbrace{\frac{Z(j_B)}{\omega}}_{<1} \right) \varepsilon_{S(j_B), \omega} \right) \quad (62)$$

where the elasticity is:

$$\varepsilon_{S(j_B), \omega} \equiv \frac{d \log S(j_B)}{d \log \omega} = \frac{1 - S(\bar{j}) - \lambda(S(j_B) - S(\bar{j}))}{\lambda S(j_B)} \left(1 - \frac{\frac{(1 - \lambda S(j_B))s(\bar{j})\omega}{S(\bar{j})(1 - S(\bar{j}) - \lambda(S(j_B) - S(\bar{j})))}}{\frac{(1 - \lambda S(j_B))s(\bar{j})\omega}{S(\bar{j})(1 - S(\bar{j}) - \lambda(S(j_B) - S(\bar{j})))}} - Z'(\bar{j}) \right)^{-1} \quad (63)$$

Proof. Consider j_B , the cutoff above which Foreign goods are boycotted. Domestic income is

$$w_H l_H = \underbrace{(1 - \lambda) \int_0^{\bar{j}} p(j) q^H(j) dj}_{\text{regular domestic households}} + \underbrace{\lambda \int_0^{j_B} p_H(j) q_H^B(j) dj}_{\text{domestic boycotters}} + \underbrace{\int_0^{\bar{j}} p(j) q^F(j) dj}_{\text{Foreign households}} \quad (64)$$

Summing the expenditures as previously, we obtain

$$w_H l_H = (1 - \lambda) S(\bar{j}) w_H l_H + \lambda S(j_B) w_H l_H + S(\bar{j}) w_F l_F \quad (65)$$

Simplifying the equation above yields:

$$\omega = \frac{S(\bar{j})}{1 - \lambda S(j_B) - (1 - \lambda) S(\bar{j})} \cdot \frac{l_F}{l_H} \quad (66)$$

From here, we see that an increase in the boycott cutoff j_B reallocates expenditure toward domestic goods and shifts the balanced trade curve (66) to the left. The equilibrium relative wage of domestic workers increases, and the trade equilibrium cutoff decreases.

The welfare of each Home boycotter is

$$U^B = \int_0^{j_B} s(j) \log(z_H(j)) dj + \int_{j_B}^1 s(j) \log(z_F(j)\omega) dj + \int_0^1 s(j) \log(s(j)l_H) dj \quad (67)$$

We obtain

$$\frac{dU^B}{dv^F} = (1 - S(j_B)) \frac{d\omega}{\omega dv^F} + s(j_B) \log \left(\underbrace{\frac{z_H(j_B)}{\omega \cdot z_F(j_B)}}_{<1} \right) \frac{dj_B}{dv^F} \quad (68)$$

Since

$$\frac{dv^F}{d\omega} = -S(\bar{j}) \frac{1}{\omega} \quad (69)$$

we obtain

$$\frac{dU^B}{dv^F} = -\frac{(1 - S(j_B))}{S(\bar{j})} - s(j_B) \log \left(\underbrace{\frac{z_H(j_B)}{\omega \cdot z_F(j_B)}}_{<1} \right) \frac{\omega \cdot dj_B}{S(\bar{j}) d\omega} \quad (70)$$

From the implicit function theorem, we have:

$$\frac{d\omega}{dj_B} = \frac{\lambda s(j_B) \omega}{1 - \lambda S(j_B) - (1 - \lambda) S(\bar{j})} \left(1 - \frac{\frac{(1 - \lambda S(j_B)) s \omega}{S(1 - \lambda S(j_B) - (1 - \lambda) S)}}{\frac{(1 - \lambda S(j_B)) s \omega}{S(1 - \lambda S(j_B) - (1 - \lambda) S)} - Z'(\bar{j})} \right) \quad (71)$$

where $S = S(\bar{j})$. □